

The fourth roots of 1 are also in the figure. They are $i, -1, -i, 1$. The angle is now $2\pi/4$ or 90° . The first root $w_4 = e^{2\pi i/4}$ is nothing but i . Even the square roots of 1 are seen, with $w_2 = e^{i2\pi/2} = -1$. Do not despise those square roots 1 and -1 . The idea behind the FFT is to go from an 8 by 8 Fourier matrix (containing powers of w_8) to the 4 by 4 matrix below (with powers of $w_4 = i$). The same idea goes from 4 to 2. By exploiting the connections of F_8 down to F_4 and up to F_{16} (and beyond), the FFT makes multiplication by F_{1024} very quick.

We describe the *Fourier matrix*, first for $n = 4$. Its rows contain powers of 1 and w and w^2 and w^3 . These are the fourth roots of 1, and their powers come in a special order.

$$\begin{array}{l} \text{Fourier} \\ \text{matrix} \\ n = 4 \end{array} \quad F = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & w & w^2 & w^3 \\ 1 & w^2 & w^4 & w^6 \\ 1 & w^3 & w^6 & w^9 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & i^2 & i^3 \\ 1 & i^2 & i^4 & i^6 \\ 1 & i^3 & i^6 & i^9 \end{bmatrix}.$$

The matrix is symmetric ($F = F^T$). It is *not* Hermitian. Its main diagonal is not real. But $\frac{1}{2}F$ is a *unitary matrix*, which means that $(\frac{1}{2}F^H)(\frac{1}{2}F) = I$:

The columns of F give $F^H F = 4I$. Its inverse is $\frac{1}{4}F^H$ which is $F^{-1} = \frac{1}{4}\overline{F}$.

The inverse changes from $w = i$ to $\overline{w} = -i$. That takes us from F to $\overline{F}$. When the Fast Fourier Transform gives a quick way to multiply by F , it does the same for F^{-1} .

The unitary matrix is $U = F/\sqrt{n}$. We avoid that $\sqrt{n}$ and just put $\frac{1}{n}$ outside F^{-1} . The main point is to multiply F times the Fourier coefficients c_0, c_1, c_2, c_3 :

$$\begin{array}{l} \text{4-point} \\ \text{Fourier} \\ \text{series} \end{array} \quad \begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \end{bmatrix} = Fc = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & w & w^2 & w^3 \\ 1 & w^2 & w^4 & w^6 \\ 1 & w^3 & w^6 & w^9 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{bmatrix}. \quad (1)$$

The input is four complex coefficients c_0, c_1, c_2, c_3 . The output is four function values y_0, y_1, y_2, y_3 . The first output $y_0 = c_0 + c_1 + c_2 + c_3$ is the value of the Fourier series at $x = 0$. The second output is the value of that series $\sum c_k e^{ikx}$ at $x = 2\pi/4$:

$$y_1 = c_0 + c_1 e^{i2\pi/4} + c_2 e^{i4\pi/4} + c_3 e^{i6\pi/4} = c_0 + c_1 w + c_2 w^2 + c_3 w^3.$$

The third and fourth outputs y_2 and y_3 are the values of $\sum c_k e^{ikx}$ at $x = 4\pi/4$ and $x = 6\pi/4$. These are *finite* Fourier series! They contain $n = 4$ terms and they are evaluated at $n = 4$ points. Those points $x = 0, 2\pi/4, 4\pi/4, 6\pi/4$ are equally spaced.

The next point would be $x = 8\pi/4$ which is 2π . Then the series is back to y_0 , because $e^{2\pi i}$ is the same as $e^0 = 1$. Everything cycles around with period 4. In this world $2 + 2$ is 0 because $(w^2)(w^2) = w^0 = 1$. We will follow the convention that j and k go from 0 to $n - 1$ (instead of 1 to n). The “zeroth row” and “zeroth column” of F contain all ones.

The n by n Fourier matrix contains powers of $w = e^{2\pi i/n}$:

$$F_n c = \begin{bmatrix} 1 & 1 & 1 & \cdot & 1 \\ 1 & w & w^2 & \cdot & w^{n-1} \\ 1 & w^2 & w^4 & \cdot & w^{2(n-1)} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ 1 & w^{n-1} & w^{2(n-1)} & \cdot & w^{(n-1)^2} \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ \cdot \\ c_{n-1} \end{bmatrix} = \begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ \cdot \\ y_{n-1} \end{bmatrix} = y. \quad (2)$$

F_n is symmetric but not Hermitian. **Its columns are orthogonal**, and $F_n \overline{F}_n = nI$. Then F_n^{-1} is $\overline{F}_n/n$. The inverse contains powers of $\overline{w}_n = e^{-2\pi i/n}$. Look at the pattern in F :

The entry in row j , column k is w^{jk} . Row zero and column zero contain $w^0 = 1$.

When we multiply c by F_n , we sum the series at n points. When we multiply y by F_n^{-1} , we find the coefficients c from the function values y . In MATLAB that command is $c = \text{fft}(y)$. The matrix F passes from “frequency space” to “physical space.”

Important note. Many authors prefer to work with $\omega = e^{-2\pi i/N}$, which is the *complex conjugate* of our w . (They often use the Greek omega, and I will do that to keep the two options separate.) With this choice, their DFT matrix contains powers of ω not w . It is $\text{conj}(F) =$ complex conjugate of our F . This takes us to frequency space.

$\overline{F}$ is a completely reasonable choice! MATLAB uses $\omega = e^{-2\pi i/N}$. The DFT matrix $\text{fft}(\text{eye}(N))$ contains powers of this number $\omega = \overline{w}$. **The Fourier matrix with w 's reconstructs y from c . The matrix $\overline{F}$ with ω 's computes Fourier coefficients as $\text{fft}(y)$.**

Also important. When a function $f(x)$ has period 2π , and we change x to $e^{i\theta}$, the function is defined around the unit circle (where $z = e^{i\theta}$). Then the Discrete Fourier Transform from y to c is matching n values of this $f(z)$ by a polynomial $p(z) = c_0 + c_1 z + \cdots + c_{n-1} z^{n-1}$.

Interpolation Find $c_0, \dots, c_{n-1}$ so that $p(z) = f(z)$ at n points $z = 1, \dots, w^{n-1}$

The Fourier matrix is the Vandermonde matrix for interpolation at those n points.

One Step of the Fast Fourier Transform

We want to multiply F times c as quickly as possible. Normally a matrix times a vector takes n^2 separate multiplications—the matrix has n^2 entries. You might think it is impossible to do better. (If the matrix has zero entries then multiplications can be skipped. But the Fourier matrix has no zeros!) By using the special pattern w^{jk} for its entries, F can be factored in a way that produces many zeros. This is the **FFT**.

The key idea is to connect F_n with the half-size Fourier matrix $F_{n/2}$. Assume that n is a power of 2 (say $n = 2^{10} = 1024$). We will connect F_{1024} to F_{512} —or rather to *two*

copies of F_{512} . When $n = 4$, the key is in the relation between these matrices:

$$F_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & i^2 & i^3 \\ 1 & i^2 & i^4 & i^6 \\ 1 & i^3 & i^6 & i^9 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} F_2 & & & \\ & F_2 & & \\ & & & \\ & & & \end{bmatrix} = \begin{bmatrix} 1 & 1 & & \\ 1 & i^2 & & \\ & & 1 & 1 \\ & & 1 & i^2 \end{bmatrix}.$$

On the left is F_4 , with no zeros. On the right is a matrix that is half zero. The work is cut in half. But wait, those matrices are not the same. We need two sparse and simple matrices to complete the FFT factorization:

Factors for FFT

$$F_4 = \begin{bmatrix} 1 & & 1 & \\ & 1 & & i \\ 1 & & -1 & \\ & 1 & & -i \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & i^2 \\ & 1 & 1 \\ & 1 & i^2 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}. \quad (3)$$

The last matrix is a permutation. It puts the even c 's (c_0 and c_2) ahead of the odd c 's (c_1 and c_3). The middle matrix performs half-size transforms F_2 and F_2 on the evens and odds. The matrix at the left combines the two half-size outputs—in a way that produces the correct full-size output $y = F_4 c$.

The same idea applies when $n = 1024$ and $m = \frac{1}{2}n = 512$. The number w is $e^{2\pi i/1024}$. It is at the angle $\theta = 2\pi/1024$ on the unit circle. The Fourier matrix F_{1024} is full of powers of w . The first stage of the FFT is the great factorization discovered by Cooley and Tukey (and foreshadowed in 1805 by Gauss):

$$F_{1024} = \begin{bmatrix} I_{512} & D_{512} \\ I_{512} & -D_{512} \end{bmatrix} \begin{bmatrix} F_{512} & \\ & F_{512} \end{bmatrix} \begin{bmatrix} \text{even-odd} \\ \text{permutation} \end{bmatrix}. \quad (4)$$

I_{512} is the identity matrix. D_{512} is the diagonal matrix with entries $(1, w, \dots, w^{511})$. The two copies of F_{512} are what we expected. Don't forget that they use the 512th root of unity (which is nothing but w^2 !!) The permutation matrix separates the incoming vector c into its even and odd parts $c' = (c_0, c_2, \dots, c_{1022})$ and $c'' = (c_1, c_3, \dots, c_{1023})$.

Here are the algebra formulas which say the same thing as the factorization of F_{1024} :

(FFT) Set $m = \frac{1}{2}n$. The first m and last m components of $y = F_n c$ combine the half-size transforms $y' = F_m c'$ and $y'' = F_m c''$. Equation (4) shows this step from n to $m = n/2$ as $I y' + D y''$ and $I y' - D y''$:

$$\begin{aligned} y_j &= y'_j + w_n^j y''_j, & j &= 0, \dots, m-1 \\ y_{j+m} &= y'_j - w_n^j y''_j, & j &= 0, \dots, m-1. \end{aligned} \quad (5)$$

Split c into c' and c'' , transform them by F_m into y' and y'' , and reconstruct y .

Those formulas come from separating even c_{2k} from odd c_{2k+1} :

$$y_j = \sum_{k=0}^{n-1} w^{jk} c_k = \sum_{k=0}^{m-1} w^{2jk} c_{2k} + \sum_{k=0}^{m-1} w^{j(2k+1)} c_{2k+1} \quad \text{with } m = \frac{1}{2}n. \quad (6)$$

The even c 's go into $c' = (c_0, c_2, \dots)$ and the odd c 's go into $c'' = (c_1, c_3, \dots)$. Then come the transforms $F_m c'$ and $F_m c''$. The key is $w_n^2 = w_m$. This gives $w_n^{2jk} = w_m^{jk}$.

Rewrite
$$y_j = \sum w_m^{jk} c'_k + (w_n)^j \sum w_m^{jk} c''_k = y'_j + (w_n)^j y''_j. \quad (7)$$

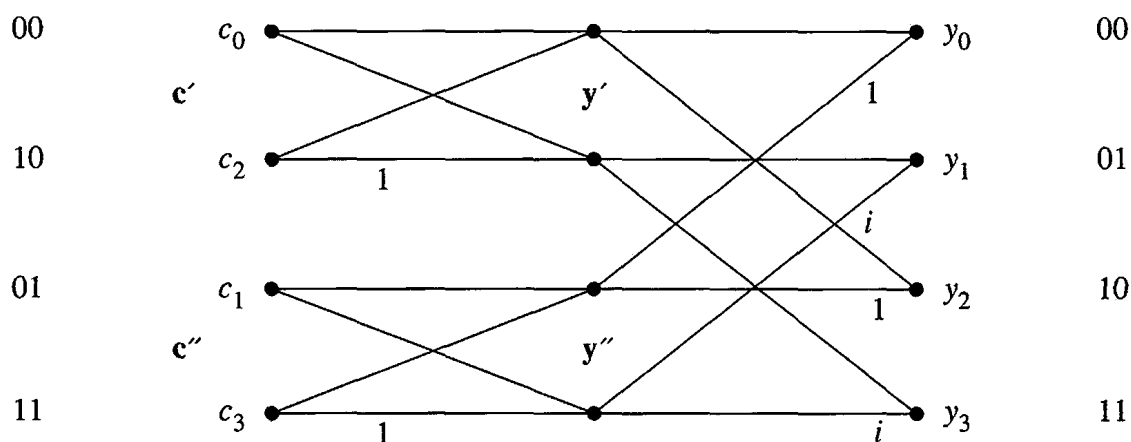
For $j \geq m$, the minus sign in (5) comes from factoring out $(w_n)^m = -1$.

MATLAB easily separates even c 's from odd c 's and multiplies by w_n^j . We use `conj(F)` or equivalently MATLAB's inverse transform `ifft`, because `fft` is based on $\omega = \bar{w} = e^{-2\pi i/n}$. Problem 17 shows that F and `conj(F)` are linked by permuting rows.

FFT step from n to $n/2$ in MATLAB	$y' = \text{ifft}(c(0:2:n-2)) * n/2;$ $y'' = \text{ifft}(c(1:2:n-1)) * n/2;$ $d = w.^{(0:n/2-1)'};$ $y = [y' + d .* y''; y' - d .* y''];$
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The flow graph shows c' and c'' going through the half-size F_2 . Those steps are called “butterflies,” from their shape. Then the outputs y' and y'' are combined (multiplying y'' by $1, i$ and also by $-1, -i$) to produce $y = F_4 c$.

This reduction from F_n to two F_m 's almost cuts the work in half—you see the zeros in the matrix factorization. That reduction is good but not great. The full idea of the **FFT** is much more powerful. It saves much more than half the time.



The Full FFT by Recursion

If you have read this far, you have probably guessed what comes next. We reduced F_n to $F_{n/2}$. **Keep going to $F_{n/4}$.** The matrices F_{512} lead to F_{256} (in four copies). Then 256 leads to 128. *That is recursion.* It is a basic principle of many fast algorithms, and here is the second stage with four copies of $F = F_{256}$ and $D = D_{256}$:

$$\begin{bmatrix} F_{512} & & \\ & F_{512} & \\ & & F_{512} & \\ & & & F_{512} \end{bmatrix} = \begin{bmatrix} I & D & & \\ I & -D & & \\ & & I & D \\ & & I & -D \end{bmatrix} \begin{bmatrix} F & & & \\ & F & & \\ & & F & \\ & & & F \end{bmatrix} \begin{bmatrix} \text{pick } 0, 4, 8, \dots \\ \text{pick } 2, 6, 10, \dots \\ \text{pick } 1, 5, 9, \dots \\ \text{pick } 3, 7, 11, \dots \end{bmatrix}.$$

We will count the individual multiplications, to see how much is saved. Before the FFT was invented, the count was the usual $n^2 = (1024)^2$. This is about a million multiplications. I am not saying that they take a long time. The cost becomes large when we have many, many transforms to do—which is typical. Then the saving by the FFT is also large:

The final count for size $n = 2^\ell$ is reduced from n^2 to $\frac{1}{2}n\ell$.

The number 1024 is 2^{10} , so $\ell = 10$. The original count of $(1024)^2$ is reduced to $(5)(1024)$. The saving is a factor of 200. A million is reduced to five thousand. That is why the FFT has revolutionized signal processing.

Here is the reasoning behind $\frac{1}{2}n\ell$. There are ℓ levels, going from $n = 2^\ell$ down to $n = 1$. Each level has $n/2$ multiplications from the diagonal D 's, to reassemble the half-size outputs from the lower level. This yields the final count $\frac{1}{2}n\ell$, which is $\frac{1}{2}n \log_2 n$.

One last note about this remarkable algorithm. There is an amazing rule for the order that the c 's enter the FFT, after all the even-odd permutations. Write the numbers 0 to $n - 1$ in binary (base 2). *Reverse the order of their digits*. The complete picture shows the bit-reversed order at the start, the $\ell = \log_2 n$ steps of the recursion, and the final output $y_0, \dots, y_{n-1}$ which is F_n times c .

The book ends with that very fundamental idea, a matrix multiplying a vector.

Thank you for studying linear algebra. I hope you enjoyed it, and I very much hope you will use it. It was a pleasure to write about this tremendously useful subject.

Problem Set 10.3

- 1 Multiply the three matrices in equation (3) and compare with F . In which six entries do you need to know that $i^2 = -1$?
- 2 Invert the three factors in equation (3) to find a fast factorization of F^{-1} .
- 3 F is symmetric. So transpose equation (3) to find a new Fast Fourier Transform!
- 4 All entries in the factorization of F_6 involve powers of $w_6 =$ sixth root of 1:

$$F_6 = \begin{bmatrix} I & D \\ I & -D \end{bmatrix} \begin{bmatrix} F_3 & \\ & F_3 \end{bmatrix} \begin{bmatrix} P \\ \end{bmatrix}.$$

Write down these matrices with $1, w_6, w_6^2$ in D and $w_3 = w_6^2$ in F_3 . Multiply!

- 5 If $v = (1, 0, 0, 0)$ and $w = (1, 1, 1, 1)$, show that $Fv = w$ and $Fw = 4v$. Therefore $F^{-1}w = v$ and $F^{-1}v = \underline{\hspace{2cm}}$.
- 6 What is F^2 and what is F^4 for the 4 by 4 Fourier matrix?
- 7 Put the vector $c = (1, 0, 1, 0)$ through the three steps of the FFT to find $y = Fc$. Do the same for $c = (0, 1, 0, 1)$.

- 8 Compute $y = F_8 c$ by the three FFT steps for $c = (1, 0, 1, 0, 1, 0, 1, 0)$. Repeat the computation for $c = (0, 1, 0, 1, 0, 1, 0, 1)$.
- 9 If $w = e^{2\pi i/64}$ then w^2 and $\sqrt{w}$ are among the _____ and _____ roots of 1.
- 10 (a) Draw all the sixth roots of 1 on the unit circle. Prove they add to zero.
(b) What are the three cube roots of 1? Do they also add to zero?
- 11 The columns of the Fourier matrix F are the *eigenvectors* of the cyclic permutation P . Multiply PF to find the eigenvalues λ_1 to λ_4 :

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & i^2 & i^3 \\ 1 & i^2 & i^4 & i^6 \\ 1 & i^3 & i^6 & i^9 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & i^2 & i^3 \\ 1 & i^2 & i^4 & i^6 \\ 1 & i^3 & i^6 & i^9 \end{bmatrix} \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \lambda_3 & \\ & & & \lambda_4 \end{bmatrix}.$$

This is $PF = F\Lambda$ or $P = F\Lambda F^{-1}$. The eigenvector matrix (usually S) is F .

- 12 The equation $\det(P - \lambda I) = 0$ is $\lambda^4 = 1$. This shows again that the eigenvalue matrix Λ is _____. Which permutation P has eigenvalues = cube roots of 1?
- 13 (a) Two eigenvectors of C are $(1, 1, 1, 1)$ and $(1, i, i^2, i^3)$. Find the eigenvalues.

$$\begin{bmatrix} c_0 & c_1 & c_2 & c_3 \\ c_3 & c_0 & c_1 & c_2 \\ c_2 & c_3 & c_0 & c_1 \\ c_1 & c_2 & c_3 & c_0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = e_1 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \quad \text{and} \quad C \begin{bmatrix} 1 \\ i \\ i^2 \\ i^3 \end{bmatrix} = e_2 \begin{bmatrix} 1 \\ i \\ i^2 \\ i^3 \end{bmatrix}.$$

(b) $P = F\Lambda F^{-1}$ immediately gives $P^2 = F\Lambda^2 F^{-1}$ and $P^3 = F\Lambda^3 F^{-1}$. Then $C = c_0 I + c_1 P + c_2 P^2 + c_3 P^3 = F(c_0 I + c_1 \Lambda + c_2 \Lambda^2 + c_3 \Lambda^3) F^{-1} = FEF^{-1}$. That matrix E in parentheses is diagonal. It contains the _____ of C .

- 14 Find the eigenvalues of the “periodic” $-1, 2, -1$ matrix from $E = 2I - \Lambda - \Lambda^3$, with the eigenvalues of P in Λ . The -1 ’s in the corners make this matrix periodic:

$$C = \begin{bmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix} \quad \text{has } c_0 = 2, c_1 = -1, c_2 = 0, c_3 = -1.$$

- 15 **Fast convolution.** To multiply C times a vector x , we can multiply $F(E(F^{-1}x))$ instead. The direct way uses n^2 separate multiplications. Knowing E and F , the second way uses only $n \log_2 n + n$ multiplications. How many of those come from E , how many from F , and how many from F^{-1} ?
- 16 Why is row i of $\overline{F}$ the same as row $N - i$ of F (numbered 0 to $N - 1$)?

Solutions to Selected Exercises

Problem Set 1.1, page 8

- 1 The combinations give (a) a line in $\mathbf{R}^3$ (b) a plane in $\mathbf{R}^3$ (c) all of $\mathbf{R}^3$.
- 4 $3\mathbf{v} + \mathbf{w} = (7, 5)$ and $c\mathbf{v} + d\mathbf{w} = (2c + d, c + 2d)$.
- 6 The components of every $c\mathbf{v} + d\mathbf{w}$ add to zero. $c = 3$ and $d = 9$ give $(3, 3, -6)$.
- 9 The fourth corner can be $(4, 4)$ or $(4, 0)$ or $(-2, 2)$.
- 11 Four more corners $(1, 1, 0)$, $(1, 0, 1)$, $(0, 1, 1)$, $(1, 1, 1)$. The center point is $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$. Centers of faces are $(\frac{1}{2}, \frac{1}{2}, 0)$, $(\frac{1}{2}, \frac{1}{2}, 1)$ and $(0, \frac{1}{2}, \frac{1}{2})$, $(1, \frac{1}{2}, \frac{1}{2})$ and $(\frac{1}{2}, 0, \frac{1}{2})$, $(\frac{1}{2}, 1, \frac{1}{2})$.
- 12 A four-dimensional cube has $2^4 = 16$ corners and $2 \cdot 4 = 8$ three-dimensional faces and 24 two-dimensional faces and 32 edges in Worked Example 2.4 A.
- 13 Sum = zero vector. Sum = $-2:00$ vector = $8:00$ vector. $2:00$ is 30° from horizontal = $(\cos \frac{\pi}{6}, \sin \frac{\pi}{6}) = (\sqrt{3}/2, 1/2)$.
- 16 All combinations with $c + d = 1$ are on the line that passes through $\mathbf{v}$ and $\mathbf{w}$. The point $\mathbf{V} = -\mathbf{v} + 2\mathbf{w}$ is on that line but it is beyond $\mathbf{w}$.
- 17 All vectors $c\mathbf{v} + d\mathbf{w}$ are on the line passing through $(0, 0)$ and $\mathbf{u} = \frac{1}{2}\mathbf{v} + \frac{1}{2}\mathbf{w}$. That line continues out beyond $\mathbf{v} + \mathbf{w}$ and back beyond $(0, 0)$. With $c \geq 0$, half of this line is removed, leaving a ray that starts at $(0, 0)$.
- 20 (a) $\frac{1}{3}\mathbf{u} + \frac{1}{3}\mathbf{v} + \frac{1}{3}\mathbf{w}$ is the center of the triangle between $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$; $\frac{1}{2}\mathbf{u} + \frac{1}{2}\mathbf{w}$ lies between $\mathbf{u}$ and $\mathbf{w}$ (b) To fill the triangle keep $c \geq 0$, $d \geq 0$, $e \geq 0$, and $c + d + e = 1$.
- 22 The vector $\frac{1}{2}(\mathbf{u} + \mathbf{v} + \mathbf{w})$ is *outside* the pyramid because $c + d + e = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} > 1$.
- 25 (a) For a line, choose $\mathbf{u} = \mathbf{v} = \mathbf{w} =$ any nonzero vector (b) For a plane, choose $\mathbf{u}$ and $\mathbf{v}$ in different directions. A combination like $\mathbf{w} = \mathbf{u} + \mathbf{v}$ is in the same plane.

Problem Set 1.2, page 19

- 3 Unit vectors $\mathbf{v}/\|\mathbf{v}\| = (\frac{3}{5}, \frac{4}{5}) = (.6, .8)$ and $\mathbf{w}/\|\mathbf{w}\| = (\frac{4}{5}, \frac{3}{5}) = (.8, .6)$. The cosine of θ is $\frac{\mathbf{v}}{\|\mathbf{v}\|} \cdot \frac{\mathbf{w}}{\|\mathbf{w}\|} = \frac{24}{25}$. The vectors $\mathbf{w}$, $\mathbf{u}$, $-\mathbf{w}$ make 0° , 90° , 180° angles with $\mathbf{w}$.
- 4 (a) $\mathbf{v} \cdot (-\mathbf{v}) = -1$ (b) $(\mathbf{v} + \mathbf{w}) \cdot (\mathbf{v} - \mathbf{w}) = \mathbf{v} \cdot \mathbf{v} + \mathbf{w} \cdot \mathbf{v} - \mathbf{v} \cdot \mathbf{w} - \mathbf{w} \cdot \mathbf{w} = 1 + () - () - 1 = 0$ so $\theta = 90^\circ$ (notice $\mathbf{v} \cdot \mathbf{w} = \mathbf{w} \cdot \mathbf{v}$) (c) $(\mathbf{v} - 2\mathbf{w}) \cdot (\mathbf{v} + 2\mathbf{w}) = \mathbf{v} \cdot \mathbf{v} - 4\mathbf{w} \cdot \mathbf{w} = 1 - 4 = -3$.

- 6 All vectors $\mathbf{w} = (c, 2c)$ are perpendicular to $\mathbf{v}$. All vectors (x, y, z) with $x + y + z = 0$ lie on a *plane*. All vectors perpendicular to $(1, 1, 1)$ and $(1, 2, 3)$ lie on a *line*.
- 9 If $v_2 w_2 / v_1 w_1 = -1$ then $v_2 w_2 = -v_1 w_1$ or $v_1 w_1 + v_2 w_2 = \mathbf{v} \cdot \mathbf{w} = 0$: perpendicular!
- 11 $\mathbf{v} \cdot \mathbf{w} < 0$ means angle $> 90^\circ$; these $\mathbf{w}$'s fill half of 3-dimensional space.
- 12 $(1, 1)$ perpendicular to $(1, 5) - c(1, 1)$ if $6 - 2c = 0$ or $c = 3$; $\mathbf{v} \cdot (\mathbf{w} - c\mathbf{v}) = 0$ if $c = \mathbf{v} \cdot \mathbf{w} / \mathbf{v} \cdot \mathbf{v}$. Subtracting $c\mathbf{v}$ is the key to perpendicular vectors.
- 15 $\frac{1}{2}(x + y) = (2 + 8)/2 = 5$; $\cos \theta = 2\sqrt{16}/\sqrt{10}\sqrt{10} = 8/10$.
- 17 $\cos \alpha = 1/\sqrt{2}$, $\cos \beta = 0$, $\cos \gamma = -1/\sqrt{2}$. For any vector $\mathbf{v}$, $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = (v_1^2 + v_2^2 + v_3^2)/\|\mathbf{v}\|^2 = 1$.
- 21 $2\mathbf{v} \cdot \mathbf{w} \leq 2\|\mathbf{v}\|\|\mathbf{w}\|$ leads to $\|\mathbf{v} + \mathbf{w}\|^2 = \mathbf{v} \cdot \mathbf{v} + 2\mathbf{v} \cdot \mathbf{w} + \mathbf{w} \cdot \mathbf{w} \leq \|\mathbf{v}\|^2 + 2\|\mathbf{v}\|\|\mathbf{w}\| + \|\mathbf{w}\|^2$. This is $(\|\mathbf{v}\| + \|\mathbf{w}\|)^2$. Taking square roots gives $\|\mathbf{v} + \mathbf{w}\| \leq \|\mathbf{v}\| + \|\mathbf{w}\|$.
- 22 $v_1^2 w_1^2 + 2v_1 w_1 v_2 w_2 + v_2^2 w_2^2 \leq v_1^2 w_1^2 + v_1^2 w_2^2 + v_2^2 w_1^2 + v_2^2 w_2^2$ is true (cancel 4 terms) because the difference is $v_1^2 w_2^2 + v_2^2 w_1^2 - 2v_1 w_1 v_2 w_2$ which is $(v_1 w_2 - v_2 w_1)^2 \geq 0$.
- 23 $\cos \beta = w_1/\|\mathbf{w}\|$ and $\sin \beta = w_2/\|\mathbf{w}\|$. Then $\cos(\beta - \alpha) = \cos \beta \cos \alpha + \sin \beta \sin \alpha = v_1 w_1 / \|\mathbf{v}\|\|\mathbf{w}\| + v_2 w_2 / \|\mathbf{v}\|\|\mathbf{w}\| = \mathbf{v} \cdot \mathbf{w} / \|\mathbf{v}\|\|\mathbf{w}\|$. This is $\cos \theta$ because $\beta - \alpha = \theta$.
- 24 Example 6 gives $|u_1||U_1| \leq \frac{1}{2}(u_1^2 + U_1^2)$ and $|u_2||U_2| \leq \frac{1}{2}(u_2^2 + U_2^2)$. The whole line becomes $.96 \leq (.6)(.8) + (.8)(.6) \leq \frac{1}{2}(.6^2 + .8^2) + \frac{1}{2}(.8^2 + .6^2) = 1$. True: $.96 < 1$.
- 28 Three vectors in the plane could make angles $> 90^\circ$ with each other: $(1, 0)$, $(-1, 4)$, $(-1, -4)$. Four vectors could not do this (360° total angle). How many can do this in $\mathbf{R}^3$ or $\mathbf{R}^n$?
- 29 Try $\mathbf{v} = (1, 2, -3)$ and $\mathbf{w} = (-3, 1, 2)$ with $\cos \theta = \frac{-7}{14}$ and $\theta = 120^\circ$. Write $\mathbf{v} \cdot \mathbf{w} = xz + yz + xy$ as $\frac{1}{2}(x + y + z)^2 - \frac{1}{2}(x^2 + y^2 + z^2)$. If $x + y + z = 0$ this is $-\frac{1}{2}(x^2 + y^2 + z^2) = -\frac{1}{2}\|\mathbf{v}\|\|\mathbf{w}\|$. Then $\mathbf{v} \cdot \mathbf{w} / \|\mathbf{v}\|\|\mathbf{w}\| = -\frac{1}{2}$.

Problem Set 1.3, page 29

- 1 $2s_1 + 3s_2 + 4s_3 = (2, 5, 9)$. The same vector $\mathbf{b}$ comes from S times $\mathbf{x} = (2, 3, 4)$:

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} (\text{row 1}) \cdot \mathbf{x} \\ (\text{row 2}) \cdot \mathbf{x} \\ (\text{row 3}) \cdot \mathbf{x} \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 9 \end{bmatrix}.$$

- 2 The solutions are $y_1 = 1$, $y_2 = 0$, $y_3 = 0$ (right side = column 1) and $y_1 = 1$, $y_2 = 3$, $y_3 = 5$. That second example illustrates that the first n odd numbers add to n^2 .
- 4 The combination $0\mathbf{w}_1 + 0\mathbf{w}_2 + 0\mathbf{w}_3$ always gives the zero vector, but this problem looks for other *zero* combinations (then the vectors are *dependent*, they lie in a plane): $\mathbf{w}_2 = (\mathbf{w}_1 + \mathbf{w}_3)/2$ so one combination that gives zero is $\frac{1}{2}\mathbf{w}_1 - \mathbf{w}_2 + \frac{1}{2}\mathbf{w}_3$.
- 5 The rows of the 3 by 3 matrix in Problem 4 must also be *dependent*: $\mathbf{r}_2 = \frac{1}{2}(\mathbf{r}_1 + \mathbf{r}_3)$. The column and row combinations that produce $\mathbf{0}$ are the same: this is unusual.
- 7 All three rows are perpendicular to the solution $\mathbf{x}$ (the three equations $\mathbf{r}_1 \cdot \mathbf{x} = 0$ and $\mathbf{r}_2 \cdot \mathbf{x} = 0$ and $\mathbf{r}_3 \cdot \mathbf{x} = 0$ tell us this). Then the whole plane of the rows is perpendicular to $\mathbf{x}$ (the plane is also perpendicular to all multiples $c\mathbf{x}$).

- 9 The cyclic difference matrix C has a line of solutions (in 4 dimensions) to $Cx = 0$:

$$\begin{bmatrix} 1 & 0 & 0 & -1 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \text{ when } x = \begin{bmatrix} c \\ c \\ c \\ c \end{bmatrix} = \text{any constant vector.}$$

- 11 The forward differences of the squares are $(t+1)^2 - t^2 = t^2 + 2t + 1 - t^2 = 2t + 1$. Differences of the n th power are $(t+1)^n - t^n = t^n - t^n + nt^{n-1} + \dots$. The leading term is the derivative nt^{n-1} . The binomial theorem gives all the terms of $(t+1)^n$.
- 12 Centered difference matrices of *even* size seem to be invertible. Look at eqns. 1 and 4:

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} \quad \begin{array}{l} \text{First} \\ \text{solve} \\ x_2 = b_1 \\ -x_3 = b_4 \end{array} \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -b_2 - b_4 \\ b_1 \\ -b_4 \\ b_1 + b_3 \end{bmatrix}$$

- 13 *Odd size*: The five centered difference equations lead to $b_1 + b_3 + b_5 = 0$.

$$\begin{array}{ll} x_2 &= b_1 \\ x_3 - x_1 &= b_2 \\ x_4 - x_2 &= b_3 \\ x_5 - x_3 &= b_4 \\ -x_4 &= b_5 \end{array} \quad \begin{array}{l} \text{Add equations 1, 3, 5} \\ \text{The left side of the sum is zero} \\ \text{The right side is } b_1 + b_3 + b_5 \\ \text{There cannot be a solution unless } b_1 + b_3 + b_5 = 0. \end{array}$$

- 14 An example is $(a, b) = (3, 6)$ and $(c, d) = (1, 2)$. The ratios a/c and b/d are equal. Then $ad = bc$. Then (when you divide by bd) the ratios a/b and c/d are equal!

Problem Set 2.1, page 40

- 1 The columns are $i = (1, 0, 0)$ and $j = (0, 1, 0)$ and $k = (0, 0, 1)$ and $b = (2, 3, 4) = 2i + 3j + 4k$.
- 2 The planes are the same: $2x = 4$ is $x = 2$, $3y = 9$ is $y = 3$, and $4z = 16$ is $z = 4$. The solution is the same point $X = x$. The columns are changed; but same combination.
- 4 If $z = 2$ then $x + y = 0$ and $x - y = z$ give the point $(1, -1, 2)$. If $z = 0$ then $x + y = 6$ and $x - y = 4$ produce $(5, 1, 0)$. Halfway between those is $(3, 0, 1)$.
- 6 Equation 1 + equation 2 - equation 3 is now $0 = -4$. Line misses plane; *no solution*.
- 8 Four planes in 4-dimensional space normally meet at a *point*. The solution to $Ax = (3, 3, 3, 2)$ is $x = (0, 0, 1, 2)$ if A has columns $(1, 0, 0, 0)$, $(1, 1, 0, 0)$, $(1, 1, 1, 0)$, $(1, 1, 1, 1)$. The equations are $x + y + z + t = 3$, $y + z + t = 3$, $z + t = 3$, $t = 2$.
- 11 Ax equals $(14, 22)$ and $(0, 0)$ and $(9, 7)$.
- 14 $2x + 3y + z + 5t = 8$ is $Ax = b$ with the 1 by 4 matrix $A = [2 \ 3 \ 1 \ 5]$. The solutions x fill a 3D "plane" in 4 dimensions. It could be called a *hyperplane*.
- 16 90° rotation from $R = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, 180° rotation from $R^2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -I$.

- 18 $E = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$ and $E = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ subtract the first component from the second.
- 22 The dot product $Ax = [1 \ 4 \ 5] \begin{bmatrix} x \\ y \\ z \end{bmatrix} = (1 \text{ by } 3)(3 \text{ by } 1)$ is zero for points (x, y, z) on a plane in three dimensions. The columns of A are one-dimensional vectors.
- 23 $A = [1 \ 2 \ ; \ 3 \ 4]$ and $x = [5 \ -2]'$ and $b = [1 \ 7]'$. $r = b - A * x$ prints as zero.
- 25 $\text{ones}(4, 4) * \text{ones}(4, 1) = [4 \ 4 \ 4 \ 4]'$; $B * w = [10 \ 10 \ 10 \ 10]'$.
- 28 The row picture shows four *lines* in the 2D plane. The column picture is in *four*-dimensional space. No solution unless the right side is a combination of *the two columns*.
- 29 u_7, v_7, w_7 are all close to $(.6, .4)$. Their components still add to 1.
- 30 $\begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix} \begin{bmatrix} .6 \\ .4 \end{bmatrix} = \begin{bmatrix} .6 \\ .4 \end{bmatrix} = \text{steady state } s$. No change when multiplied by $\begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix}$.
- 31 $M = \begin{bmatrix} 8 & 3 & 4 \\ 1 & 5 & 9 \\ 6 & 7 & 2 \end{bmatrix} = \begin{bmatrix} 5+u & 5-u+v & 5-v \\ 5-u-v & 5 & 5+u+v \\ 5+v & 5+u-v & 5-u \end{bmatrix}$; $M_3(1, 1, 1) = (15, 15, 15)$;
 $M_4(1, 1, 1, 1) = (34, 34, 34, 34)$ because $1 + 2 + \cdots + 16 = 136$ which is $4(34)$.
- 32 A is singular when its third column w is a combination $cu + dv$ of the first columns. A typical column picture has b outside the plane of u, v, w . A typical row picture has the intersection line of two planes parallel to the third plane. *Then no solution.*
- 33 $w = (5, 7)$ is $5u + 7v$. Then Aw equals 5 times Au plus 7 times Av .
- 34 $\begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$ has the solution $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \\ 8 \\ 6 \end{bmatrix}$.
- 35 $x = (1, \dots, 1)$ gives $Sx = \text{sum of each row} = 1 + \cdots + 9 = 45$ for Sudoku matrices. 6 row orders $(1, 2, 3), (1, 3, 2), (2, 1, 3), (2, 3, 1), (3, 1, 2), (3, 2, 1)$ are in Section 2.7. The same 6 permutations of *blocks* of rows produce Sudoku matrices, so $6^4 = 1296$ orders of the 9 rows all stay Sudoku. (And also 1296 permutations of the 9 columns.)

Problem Set 2.2, page 51

- 3 Subtract $-\frac{1}{2}$ (or add $\frac{1}{2}$) times equation 1. The new second equation is $3y = 3$. Then $y = 1$ and $x = 5$. If the right side changes sign, so does the solution: $(x, y) = (-5, -1)$.
- 4 Subtract $\ell = \frac{c}{a}$ times equation 1. The new second pivot multiplying y is $d - (cb/a)$ or $(ad - bc)/a$. Then $y = (ag - cf)/(ad - bc)$.
- 6 Singular system if $b = 4$, because $4x + 8y$ is 2 times $2x + 4y$. Then $g = 32$ makes the lines become the *same*: infinitely many solutions like $(8, 0)$ and $(0, 4)$.
- 8 If $k = 3$ elimination must fail: no solution. If $k = -3$, elimination gives $0 = 0$ in equation 2: infinitely many solutions. If $k = 0$ a row exchange is needed: one solution.
- 14 Subtract 2 times row 1 from row 2 to reach $(d - 10)y - z = 2$. Equation (3) is $y - z = 3$. If $d = 10$ exchange rows 2 and 3. If $d = 11$ the system becomes singular.

- 15 The second pivot position will contain $-2 - b$. If $b = -2$ we exchange with row 3. If $b = -1$ (singular case) the second equation is $-y - z = 0$. A solution is $(1, 1, -1)$.
- 17 If row 1 = row 2, then row 2 is zero after the first step; exchange the zero row with row 3 and there is no *third* pivot. If column 2 = column 1, then column 2 has no pivot.
- 19 Row 2 becomes $3y - 4z = 5$, then row 3 becomes $(q + 4)z = t - 5$. If $q = -4$ the system is singular — no third pivot. Then if $t = 5$ the third equation is $0 = 0$. Choosing $z = 1$ the equation $3y - 4z = 5$ gives $y = 3$ and equation 1 gives $x = -9$.
- 20 Singular if row 3 is a combination of rows 1 and 2. From the end view, the three planes form a triangle. This happens if rows $1 + 2 = \text{row 3}$ on the left side but not the right side: $x + y + z = 0$, $x - 2y - z = 1$, $2x - y = 4$. No parallel planes but still no solution.
- 25 $a = 2$ (equal columns), $a = 4$ (equal rows), $a = 0$ (zero column).
- 28 $A(2, :) = A(2, :) - 3 * A(1, :)$ will subtract 3 times row 1 from row 2.
- 29 Pivots 2 and 3 can be arbitrarily large. I believe their averages are infinite! *With row exchanges* in MATLAB's `lu` code, the averages are much more stable (and should be predictable, also for `randn` with normal instead of uniform probability distribution).
- 30 If $A(5, 5)$ is 7 not 11, then the last pivot will be 0 not 4.
- 31 Row j of U is a combination of rows $1, \dots, j$ of A . If $Ax = 0$ then $Ux = 0$ (not true if b replaces 0). U is the diagonal of A when A is *lower triangular*.

Problem Set 2.3, page 63

- 1 $E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 7 & 1 \end{bmatrix}$, $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$.
- 3 $\begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}$ $M = E_{32}E_{31}E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 10 & -2 & 1 \end{bmatrix}$.
- 5 Changing a_{33} from 7 to 11 will change the third pivot from 5 to 9. Changing a_{33} from 7 to 2 will change the pivot from 5 to *no pivot*.
- 9 $M = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$. After the exchange, we need E_{31} (not E_{21}) to act on the new row 3.
- 10 $E_{13} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}; \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}; E_{31}E_{13} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$. Test on the identity matrix!
- 12 The first product is $\begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}$ rows and also columns reversed. The second product is $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -2 \\ 0 & 2 & -3 \end{bmatrix}$.
- 14 E_{21} has $-\ell_{21} = \frac{1}{2}$, E_{32} has $-\ell_{32} = \frac{2}{3}$, E_{43} has $-\ell_{43} = \frac{3}{4}$. Otherwise the E 's match I .
- 18 $EF = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix}$, $FE = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b+ac & c & 1 \end{bmatrix}$, $E^2 = \begin{bmatrix} 1 & 0 & 0 \\ 2a & 1 & 0 \\ 2b & 0 & 1 \end{bmatrix}$, $F^3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3c & 1 \end{bmatrix}$.

- 22 (a) $\sum a_{3j}x_j$ (b) $a_{21}-a_{11}$ (c) $a_{21}-2a_{11}$ (d) $(E_{21}Ax)_1 = (Ax)_1 = \sum a_{1j}x_j$.
- 25 The last equation becomes $0 = 3$. If the original 6 is 3, then row 1 + row 2 = row 3.
- 27 (a) No solution if $d=0$ and $c \neq 0$ (b) Many solutions if $d=0=c$. No effect from a, b .
- 28 $A = AI = A(BC) = (AB)C = IC = C$. That middle equation is crucial.
- 30 $EM = \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}$ then $FEM = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$ then $EFEM = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$ then $EEFEM = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = B$. So after inverting with $E^{-1} = A$ and $F^{-1} = B$ this is $M = ABAAB$.

Problem Set 2.4, page 75

- 2 (a) A (column 3 of B) (b) (Row 1 of A) B (c) (Row 3 of A)(column 4 of B)
(d) (Row 1 of C) D (column 1 of E).
- 5 (a) $A^2 = \begin{bmatrix} 1 & 2b \\ 0 & 1 \end{bmatrix}$ and $A^n = \begin{bmatrix} 1 & nb \\ 0 & 1 \end{bmatrix}$. (b) $A^2 = \begin{bmatrix} 4 & 4 \\ 0 & 0 \end{bmatrix}$ and $A^n = \begin{bmatrix} 2^n & 2^n \\ 0 & 0 \end{bmatrix}$.
- 7 (a) True (b) False (c) True (d) False.
- 9 $AF = \begin{bmatrix} a & a+b \\ c & c+d \end{bmatrix}$ and $E(AF) = (EA)F$: Matrix multiplication is *associative*.
- 11 (a) $B = 4I$ (b) $B = 0$ (c) $B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ (d) Every row of B is 1, 0, 0.
- 15 (a) mn (use every entry of A) (b) $mnp = p \times \text{part (a)}$ (c) n^3 (n^2 dot products).
- 16 (a) Use only column 2 of B (b) Use only row 2 of A (c)–(d) Use row 2 of first A .
- 18 Diagonal matrix, lower triangular, symmetric, all rows equal. Zero matrix fits all four.
- 19 (a) a_{11} (b) $\ell_{31} = a_{31}/a_{11}$ (c) $a_{32} - (\frac{a_{31}}{a_{11}})a_{12}$ (d) $a_{22} - (\frac{a_{21}}{a_{11}})a_{12}$.
- 22 $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ has $A^2 = -I$; $BC = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$;
 $DE = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = -ED$. You can find more examples.
- 24 $(A_1)^n = \begin{bmatrix} 2^n & 2^n - 1 \\ 0 & 1 \end{bmatrix}$, $(A_2)^n = 2^{n-1} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, $(A_3)^n = \begin{bmatrix} a^n & a^{n-1}b \\ 0 & 0 \end{bmatrix}$.
- 27 (a) (row 3 of A) $\cdot$ (column 1 of B) and (row 3 of A) $\cdot$ (column 2 of B) are both zero.
(b) $\begin{bmatrix} x \\ x \\ 0 \end{bmatrix} \begin{bmatrix} 0 & x & x \end{bmatrix} = \begin{bmatrix} 0 & x & x \\ 0 & x & x \\ 0 & 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} x \\ x \\ x \end{bmatrix} \begin{bmatrix} 0 & 0 & x \end{bmatrix} = \begin{bmatrix} 0 & 0 & x \\ 0 & 0 & x \\ 0 & 0 & x \end{bmatrix}$: **both upper**.
- 28 A times B with cuts $A \begin{bmatrix} | & | & | & | \end{bmatrix}$, $\begin{bmatrix} \text{---} \end{bmatrix} B$, $\begin{bmatrix} \text{---} \end{bmatrix} \begin{bmatrix} | & | & | & | \end{bmatrix}$, $\begin{bmatrix} | & | & | \end{bmatrix} \begin{bmatrix} \text{---} \end{bmatrix}$
- 30 In 29, $c = \begin{bmatrix} -2 \\ 8 \end{bmatrix}$, $D = \begin{bmatrix} 0 & 1 \\ 5 & 3 \end{bmatrix}$, $D - cb/a = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}$ in the lower corner of EA .
- 32 A times $X = [x_1 \ x_2 \ x_3]$ will be the identity matrix $I = [Ax_1 \ Ax_2 \ Ax_3]$.

33 $b = \begin{bmatrix} 3 \\ 5 \\ 8 \end{bmatrix}$ gives $x = 3x_1 + 5x_2 + 8x_3 = \begin{bmatrix} 3 \\ 8 \\ 16 \end{bmatrix}$; $A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$ will have

those $x_1 = (1, 1, 1)$, $x_2 = (0, 1, 1)$, $x_3 = (0, 0, 1)$ as columns of its “inverse” A^{-1} .

35 $A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$, $A^2 = \begin{bmatrix} 2 & 0 & 2 & 0 \\ 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 0 \\ 0 & 2 & 0 & 2 \end{bmatrix}$, **aba, ada cba, cda** These show
bab, bcb dab, dcb 16 2-step
abc, adc cbc, cdc paths in
bad, bcd dad, dcd the graph

Problem Set 2.5, page 89

1 $A^{-1} = \begin{bmatrix} 0 & \frac{1}{4} \\ \frac{1}{3} & 0 \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ -1 & \frac{1}{2} \end{bmatrix}$ and $C^{-1} = \begin{bmatrix} 7 & -4 \\ -5 & 3 \end{bmatrix}$.

7 (a) In $Ax = (1, 0, 0)$, equation 1 + equation 2 – equation 3 is $0 = 1$ (b) Right sides must satisfy $b_1 + b_2 = b_3$ (c) Row 3 becomes a row of zeros—no third pivot.

8 (a) The vector $x = (1, 1, -1)$ solves $Ax = 0$ (b) After elimination, columns 1 and 2 end in zeros. Then so does column 3 = column 1 + 2: no third pivot.

12 Multiply $C = AB$ on the left by A^{-1} and on the right by C^{-1} . Then $A^{-1} = BC^{-1}$.

14 $B^{-1} = A^{-1} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}^{-1} = A^{-1} \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$: subtract column 2 of A^{-1} from column 1.

16 $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} ad - bc & 0 \\ 0 & ad - bc \end{bmatrix}$. The inverse of each matrix is the other divided by $ad - bc$.

18 $A^2B = I$ can also be written as $A(AB) = I$. Therefore A^{-1} is AB .

21 Six of the sixteen 0 – 1 matrices are invertible, including all four with three 1's.

22 $\begin{bmatrix} 1 & 3 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 7 & -3 \\ 0 & 1 & -2 & 1 \end{bmatrix} = [I \ A^{-1}]$;
 $\begin{bmatrix} 1 & 4 & 1 & 0 \\ 3 & 9 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 4 & 1 & 0 \\ 0 & -3 & -3 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -3 & 4/3 \\ 0 & 1 & 1 & -1/3 \end{bmatrix} = [I \ A^{-1}]$.

24 $\begin{bmatrix} 1 & a & b & 1 & 0 & 0 \\ 0 & 1 & c & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & a & 0 & 1 & 0 & -b \\ 0 & 1 & 0 & 0 & 1 & -c \\ 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 1 & -a & ac - b \\ 0 & 1 & 0 & 0 & 1 & -c \\ 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$.

27 $A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}$ (notice the pattern); $A^{-1} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$.

31 Elimination produces the pivots a and $a - b$ and $a - b$. $A^{-1} = \frac{1}{a(a - b)} \begin{bmatrix} a & 0 & -b \\ -a & a & 0 \\ 0 & -a & a \end{bmatrix}$.

33 $x = (1, 1, \dots, 1)$ has $Px = Qx$ so $(P - Q)x = 0$.

34 $\begin{bmatrix} I & 0 \\ -C & I \end{bmatrix}$ and $\begin{bmatrix} A^{-1} & 0 \\ -D^{-1}CA^{-1} & D^{-1} \end{bmatrix}$ and $\begin{bmatrix} -D & I \\ I & 0 \end{bmatrix}$.

35 A can be invertible with diagonal zeros. B is singular because each row adds to zero.

- 38 The three Pascal matrices have $P = LU = LL^T$ and then $\text{inv}(P) = \text{inv}(L^T)\text{inv}(L)$.
- 42 $MM^{-1} = (I_n - UV)(I_n + U(I_m - VU)^{-1}V)$ (this is testing formula 3)
 $= I_n - UV + U(I_m - VU)^{-1}V - UVU(I_m - VU)^{-1}V$ (keep simplifying)
 $= I_n - UV + U(I_m - VU)(I_m - VU)^{-1}V = I_n$ (formulas 1, 2, 4 are similar)
- 43 4 by 4 still with $T_{11} = 1$ has pivots 1, 1, 1, 1; reversing to $T^* = UL$ makes $T_{44}^* = 1$.
- 44 Add the equations $Cx = b$ to find $0 = b_1 + b_2 + b_3 + b_4$. Same for $Fx = b$.

Problem Set 2.6, page 102

- 3 $\ell_{31} = 1$ and $\ell_{32} = 2$ (and $\ell_{33} = 1$): reverse steps to get $Au = b$ from $Ux = c$:
 1 times $(x + y + z = 5) + 2$ times $(y + 2z = 2) + 1$ times $(z = 2)$ gives $x + 3y + 6z = 11$.
- 4 $Lc = \begin{bmatrix} 1 & & \\ 1 & 1 & \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ 11 \end{bmatrix}$; $Ux = \begin{bmatrix} 1 & 1 & 1 \\ & 1 & 2 \\ & & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ 2 \end{bmatrix}$; $x = \begin{bmatrix} 5 \\ -2 \\ 2 \end{bmatrix}$.
- 6 $\begin{bmatrix} 1 & & \\ 0 & 1 & \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ -2 & 1 & \\ 0 & 0 & 1 \end{bmatrix} A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & -6 \end{bmatrix} = U$. Then $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix}$ U is the same as $E_{21}^{-1}E_{32}^{-1}U = LU$. The multipliers $\ell_{21}, \ell_{32} = 2$ fall into place in L .
- 10 $c = 2$ leads to zero in the second pivot position: exchange rows and not singular.
 $c = 1$ leads to zero in the third pivot position. In this case the matrix is *singular*.
- 12 $A = \begin{bmatrix} 2 & 4 \\ 4 & 11 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = LDU$; U is L^T
 $\begin{bmatrix} 1 & & \\ 4 & 1 & \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 4 & 0 \\ 0 & -4 & 4 \\ 0 & 0 & 4 \end{bmatrix} = \begin{bmatrix} 1 & & \\ 4 & 1 & \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ & -4 & \\ & & 4 \end{bmatrix} \begin{bmatrix} 1 & 4 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} = LDL^T$.
- 14 $\begin{bmatrix} a & r & r & r \\ a & b & s & s \\ a & b & c & t \\ a & b & c & d \end{bmatrix} = \begin{bmatrix} 1 & & & \\ 1 & 1 & & \\ 1 & 1 & 1 & \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} a & r & r & r \\ & b-r & s-r & s-r \\ & & c-s & t-s \\ & & & d-t \end{bmatrix}$. Need $\begin{matrix} a \neq 0 \\ b \neq r \\ c \neq s \\ d \neq t \end{matrix}$
- 15 $\begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} c = \begin{bmatrix} 2 \\ 11 \end{bmatrix}$ gives $c = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$. Then $\begin{bmatrix} 2 & 4 \\ 0 & 1 \end{bmatrix} x = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ gives $x = \begin{bmatrix} -5 \\ 3 \end{bmatrix}$.
 $Ax = b$ is $LUx = \begin{bmatrix} 2 & 4 \\ 8 & 17 \end{bmatrix} x = \begin{bmatrix} 2 \\ 11 \end{bmatrix}$. Forward to $\begin{bmatrix} 2 & 4 \\ 0 & 1 \end{bmatrix} x = \begin{bmatrix} 2 \\ 3 \end{bmatrix} = c$.
- 18 (a) Multiply $LDU = L_1 D_1 U_1$ by inverses to get $L_1^{-1}LD = D_1 U_1 U^{-1}$. The left side is lower triangular, the right side is upper triangular $\Rightarrow$ both sides are diagonal.
 (b) L, U, L_1, U_1 have diagonal 1's so $D = D_1$. Then $L_1^{-1}L$ and $U_1 U^{-1}$ are both I .
- 20 A tridiagonal T has 2 nonzeros in the pivot row and only one nonzero below the pivot (one operation to find ℓ and then one for the new pivot!). $T = \text{bidiagonal } L \text{ times bidiagonal } U$.
- 23 The 2 by 2 upper submatrix A_2 has the first two pivots 5, 9. Reason: Elimination on A starts in the upper left corner with elimination on A_2 .
- 24 The upper left blocks all factor at the same time as A : A_k is $L_k U_k$.
- 25 The i, j entry of L^{-1} is j/i for $i \geq j$. And $L_{i,i-1}$ is $(1-i)/i$ below the diagonal
- 26 $(K^{-1})_{ij} = j(n-i+1)/(n+1)$ for $i \geq j$ (and symmetric): $(n+1)K^{-1}$ looks good.

Problem Set 2.7, page 115

- 2 $(AB)^T$ is not $A^T B^T$ except when $AB = BA$. Transpose that to find: $B^T A^T = A^T B^T$.
- 4 $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ has $A^2 = 0$. The diagonal of $A^T A$ has dot products of columns of A with themselves. If $A^T A = 0$, zero dot products $\Rightarrow$ zero columns $\Rightarrow A =$ zero matrix.
- 6 $M^T = \begin{bmatrix} A^T & C^T \\ B^T & D^T \end{bmatrix}$; $M^T = M$ needs $A^T = A$ and $B^T = C$ and $D^T = D$.
- 8 The 1 in row 1 has n choices; then the 1 in row 2 has $n - 1$ choices ... ($n!$ overall).
- 10 (3, 1, 2, 4) and (2, 3, 1, 4) keep 4 in place; 6 more even P 's keep 1 or 2 or 3 in place; (2, 1, 4, 3) and (3, 4, 1, 2) exchange 2 pairs. (1, 2, 3, 4), (4, 3, 2, 1) make 12 even P 's.
- 14 The i, j entry of PAP is the $n - i + 1, n - j + 1$ entry of A . Diagonal will reverse order.
- 18 (a) $5 + 4 + 3 + 2 + 1 = 15$ independent entries if $A = A^T$ (b) L has 10 and D has 5; total 15 in LDL^T (c) Zero diagonal if $A^T = -A$, leaving $4 + 3 + 2 + 1 = 10$ choices.
- 20 $\begin{bmatrix} 1 & 3 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -7 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$; $\begin{bmatrix} 1 & b \\ b & c \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ b & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & c - b^2 \end{bmatrix} \begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix}$
 $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & & \\ -\frac{1}{2} & 1 & \\ 0 & -\frac{2}{3} & 1 \end{bmatrix} \begin{bmatrix} 2 & & \\ & \frac{3}{2} & \\ & & \frac{4}{3} \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{2} & 0 \\ & 1 & -\frac{2}{3} \\ & & 1 \end{bmatrix} = LDL^T.$
- 22 $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} A = \begin{bmatrix} 1 & & \\ 0 & 1 & \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ & 1 & 1 \\ & & -1 \end{bmatrix}$; $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} A = \begin{bmatrix} 1 & & \\ 1 & 1 & \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ & -1 & 1 \\ & & 1 \end{bmatrix}$
- 24 $PA = LU$ is $\begin{bmatrix} & & 1 \\ & 1 & \\ 1 & & \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 \\ 0 & 3 & 8 \\ 2 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & & \\ 0 & 1 & \\ 0 & 1/3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ & 3 & 8 \\ & & -2/3 \end{bmatrix}$. If we wait to exchange and a_{12} is the pivot, $A = L_1 P_1 U_1 = \begin{bmatrix} 1 & & \\ 3 & 1 & \\ & & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ & 1 & \\ 1 & & \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix}$.
- 26 One way to decide even vs. odd is to count all pairs that P has in the wrong order. Then P is even or odd when that count is even or odd. Hard step: Show that an exchange always switches that count! Then 3 or 5 exchanges will leave that count odd.
- 31 $\begin{bmatrix} 1 & 50 \\ 40 & 1000 \\ 2 & 50 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = Ax$; $A^T y = \begin{bmatrix} 1 & 40 & 2 \\ 50 & 1000 & 50 \end{bmatrix} \begin{bmatrix} 700 \\ 3 \\ 3000 \end{bmatrix} = \begin{bmatrix} 6820 \\ 188000 \end{bmatrix}$ 1 truck
1 plane
- 32 $Ax \cdot y$ is the cost of inputs while $x \cdot A^T y$ is the value of outputs.
- 33 $P^3 = I$ so three rotations for 360° ; P rotates around (1, 1, 1) by 120° .
- 36 These are groups: Lower triangular with diagonal 1's, diagonal invertible D , permutations P , orthogonal matrices with $Q^T = Q^{-1}$.
- 37 Certainly B^T is northwest. B^2 is a full matrix! B^{-1} is southeast: $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}$. The rows of B are in reverse order from a lower triangular L , so $B = PL$. Then $B^{-1} = L^{-1}P^{-1}$ has the columns in reverse order from L^{-1} . So B^{-1} is southeast. Northwest $B = PL$ times southeast PU is $(PLP)U =$ upper triangular.

- 38** There are $n!$ permutation matrices of order n . Eventually *two* powers of P must be the same: If $P^r = P^s$ then $P^{r-s} = I$. Certainly $r - s \leq n!$

$$P = \begin{bmatrix} P_2 & \\ & P_3 \end{bmatrix} \text{ is 5 by 5 with } P_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \text{ and } P_3 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \text{ and } P^6 = I.$$

Problem Set 3.1, page 127

- 1** $x + y \neq y + x$ and $x + (y + z) \neq (x + y) + z$ and $(c_1 + c_2)x \neq c_1x + c_2x$.
- 3** (a) cx may not be in our set: not closed under multiplication. Also no 0 and no $-x$
 (b) $c(x + y)$ is the usual $(xy)^c$, while $cx + cy$ is the usual $(x^c)(y^c)$. Those are equal. With $c = 3$, $x = 2$, $y = 1$ this is $3(2 + 1) = 8$. The zero vector is the number 1.
- 5** (a) One possibility: The matrices cA form a subspace not containing B (b) Yes: the subspace must contain $A - B = I$ (c) Matrices whose main diagonal is all zero.
- 9** (a) The vectors with integer components allow addition, but not multiplication by $\frac{1}{2}$
 (b) Remove the x axis from the xy plane (but leave the origin). Multiplication by any c is allowed but not all vector additions.
- 11** (a) All matrices $\begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix}$ (b) All matrices $\begin{bmatrix} a & a \\ 0 & 0 \end{bmatrix}$ (c) All diagonal matrices.
- 15** (a) Two planes through $(0, 0, 0)$ probably intersect in a line through $(0, 0, 0)$
 (b) The plane and line probably intersect in the point $(0, 0, 0)$
 (c) If x and y are in both S and T , $x + y$ and cx are in both subspaces.
- 20** (a) Solution only if $b_2 = 2b_1$ and $b_3 = -b_1$ (b) Solution only if $b_3 = -b_1$.
- 23** The extra column b enlarges the column space unless b is *already* in the column space.
 $[A \ b] = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ (larger column space) $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ (b is in column space)
 (no solution to $Ax = b$) ($Ax = b$ has a solution)
- 25** The solution to $Az = b + b^*$ is $z = x + y$. If b and b^* are in $C(A)$ so is $b + b^*$.
- 30** (a) If u and v are both in $S + T$, then $u = s_1 + t_1$ and $v = s_2 + t_2$. So $u + v = (s_1 + s_2) + (t_1 + t_2)$ is also in $S + T$. And so is $cu = cs_1 + ct_1$: a subspace.
 (b) If S and T are different lines, then $S \cup T$ is just the two lines (*not a subspace*) but $S + T$ is the whole plane that they span.
- 31** If $S = C(A)$ and $T = C(B)$ then $S + T$ is the column space of $M = [A \ B]$.
- 32** The columns of AB are combinations of the columns of A . So all columns of $[A \ AB]$ are already in $C(A)$. But $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ has a larger column space than $A^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.
 For square matrices, the column space is $\mathbf{R}^n$ when A is invertible.

Problem Set 3.2, page 140

- 2** (a) Free variables x_2, x_4, x_5 and solutions $(-2, 1, 0, 0, 0)$, $(0, 0, -2, 1, 0)$, $(0, 0, -3, 0, 1)$
 (b) Free variable x_3 : solution $(1, -1, 1)$. Special solution for each free variable.

- 4 $R = \begin{bmatrix} 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$, $R = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, R has the same nullspace as U and A .
- 6 (a) Special solutions $(3, 1, 0)$ and $(5, 0, 1)$ (b) $(3, 1, 0)$. Total of pivot and free is n .
- 8 $R = \begin{bmatrix} 1 & -3 & -5 \\ 0 & 0 & 0 \end{bmatrix}$ with $I = [1]$; $R = \begin{bmatrix} 1 & -3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ with $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.
- 10 (a) Impossible row 1 (b) A = invertible (c) A = all ones (d) $A = 2I$, $R = I$.
- 14 If column 1 = column 5 then x_5 is a free variable. Its special solution is $(-1, 0, 0, 0, 1)$.
- 16 The nullspace contains only $x = 0$ when A has 5 pivots. Also the column space is $\mathbf{R}^5$, because we can solve $Ax = b$ and every b is in the column space.
- 20 Column 5 is sure to have no pivot since it is a combination of earlier columns. With 4 pivots in the other columns, the special solution is $s = (1, 0, 1, 0, 1)$. The nullspace contains all multiples of this vector s (a line in $\mathbf{R}^5$).
- 24 This construction is impossible: 2 pivot columns and 2 free variables, only 3 columns.
- 26 $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ has $N(A) = C(A)$ and also (a)(b)(c) are all false. Notice $\text{rref}(A^T) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.
- 30
- 32 Any zero rows come after these rows: $R = [1 \ -2 \ -3]$, $R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, $R = I$.
- 33 (a) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ (b) All 8 matrices are R 's!
- 35 The nullspace of $B = [A \ A]$ contains all vectors $x = \begin{bmatrix} y \\ -y \end{bmatrix}$ for y in $\mathbf{R}^4$.
- 36 If $Cx = 0$ then $Ax = 0$ and $Bx = 0$. So $N(C) = N(A) \cap N(B) = \text{intersection}$.
- 37 *Currents*: $y_1 - y_3 + y_4 = -y_1 + y_2 + y_5 = -y_2 + y_4 + y_6 = -y_4 - y_5 - y_6 = 0$. These equations add to $0 = 0$. Free variables y_3, y_5, y_6 : watch for flows around loops.

Problem Set 3.3, page 151

- 1 (a) and (c) are correct; (d) is false because R might have 1's in nonpivot columns.
- 3 $R_A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ $R_B = [R_A \ R_A]$ $R_C \rightarrow \begin{bmatrix} R_A & 0 \\ 0 & R_A \end{bmatrix} \rightarrow$ Zero rows go to the bottom
- 5 I think $R_1 = A_1$, $R_2 = A_2$ is true. But $R_1 - R_2$ may have -1 's in some pivots.
- 7 Special solutions in $N = [-2 \ -4 \ 1 \ 0; -3 \ -5 \ 0 \ 1]$ and $[1 \ 0 \ 0; 0 \ -2 \ 1]$.
- 13 P has rank r (the same as A) because elimination produces the same pivot columns.
- 14 The rank of R^T is also r . The example matrix A has rank 2 with invertible S :

$$P = \begin{bmatrix} 1 & 3 \\ 2 & 6 \\ 2 & 7 \end{bmatrix} \quad P^T = \begin{bmatrix} 1 & 2 & 2 \\ 3 & 6 & 7 \end{bmatrix} \quad S^T = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix} \quad S = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}.$$

- 16 $(uv^T)(wz^T) = u(v^Tw)z^T$ has rank one unless the inner product is $v^Tw = 0$.

- 18 If we know that $\text{rank}(B^T A^T) \leq \text{rank}(A^T)$, then since rank stays the same for transposes, (apologies that this fact is not yet proved), we have $\text{rank}(AB) \leq \text{rank}(A)$.
- 20 Certainly A and B have at most rank 2. Then their product AB has at most rank 2. Since BA is 3 by 3, it cannot be I even if $AB = I$.
- 21 (a) A and B will both have the same nullspace and row space as the R they share.
 (b) A equals an invertible matrix times B , when they share the same R . A key fact!
- 22 $A = (\text{pivot columns})(\text{nonzero rows of } R) = \begin{bmatrix} 1 & 0 \\ 1 & 4 \\ 1 & 8 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 8 \end{bmatrix}$.
 $B = \begin{bmatrix} 2 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{matrix} \text{columns} \\ \text{times rows} \end{matrix} = \begin{bmatrix} 2 & 0 \\ 2 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ 0 & 3 \end{bmatrix}$
- 26 The m by n matrix Z has r ones to start its main diagonal. Otherwise Z is all zeros.
- 27 $R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} r \text{ by } r & r \text{ by } n-r \\ m-r \text{ by } r & m-r \text{ by } n-r \end{bmatrix}$; $\text{rref}(R^T) = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix}$; $\text{rref}(R^T R) = \text{same } R$
- 28 The row-column reduced echelon form is always $\begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix}$; I is r by r .

Problem Set 3.4, page 163

- 2 $\begin{bmatrix} 2 & 1 & 3 & \mathbf{b}_1 \\ 6 & 3 & 9 & \mathbf{b}_2 \\ 4 & 2 & 6 & \mathbf{b}_3 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 & 3 & \mathbf{b}_1 \\ 0 & 0 & 0 & \mathbf{b}_2 - 3\mathbf{b}_1 \\ 0 & 0 & 0 & \mathbf{b}_3 - 2\mathbf{b}_1 \end{bmatrix}$ Then $[R \ d] = \begin{bmatrix} 1 & 1/2 & 3/2 & 5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$
 $Ax = b$ has a solution when $b_2 - 3b_1 = 0$ and $b_3 - 2b_1 = 0$; $C(A) =$ line through $(2, 6, 4)$ which is the intersection of the planes $b_2 - 3b_1 = 0$ and $b_3 - 2b_1 = 0$; the nullspace contains all combinations of $s_1 = (-1/2, 1, 0)$ and $s_2 = (-3/2, 0, 1)$; particular solution $x_p = d = (5, 0, 0)$ and complete solution $x_p + c_1 s_1 + c_2 s_2$.
- 4 $x_{\text{complete}} = x_p + x_n = (\frac{1}{2}, 0, \frac{1}{2}, 0) + x_2(-3, 1, 0, 0) + x_4(0, 0, -2, 1)$.
- 6 (a) Solvable if $b_2 = 2b_1$ and $3b_1 - 3b_3 + b_4 = 0$. Then $x = \begin{bmatrix} 5b_1 - 2b_3 \\ b_3 - 2b_1 \end{bmatrix} = x_p$
 (b) Solvable if $b_2 = 2b_1$ and $3b_1 - 3b_3 + b_4 = 0$. $x = \begin{bmatrix} 5b_1 - 2b_3 \\ b_3 - 2b_1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$.
- 8 (a) Every b is in $C(A)$: independent rows, only the zero combination gives 0 .
 (b) Need $b_3 = 2b_2$, because $(\text{row } 3) - 2(\text{row } 2) = 0$.
- 12 (a) $x_1 - x_2$ and 0 solve $Ax = 0$ (b) $A(2x_1 - 2x_2) = 0$, $A(2x_1 - x_2) = b$
- 13 (a) The particular solution x_p is always multiplied by 1 (b) Any solution can be x_p
 (c) $\begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \end{bmatrix}$. Then $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is shorter (length $\sqrt{2}$) than $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$ (length 2)
 (d) The only "homogeneous" solution in the nullspace is $x_n = 0$ when A is invertible.
- 14 If column 5 has no pivot, x_5 is a free variable. The zero vector is not the only solution to $Ax = 0$. If this system $Ax = b$ has a solution, it has infinitely many solutions.

- 16 The largest rank is 3. Then there is a pivot in every row. The solution *always exists*. The column space is $\mathbf{R}^3$. An example is $A = [I \ F]$ for any 3 by 2 matrix F .
- 18 Rank = 2; rank = 3 unless $q = 2$ (then rank = 2). Transpose has the same rank!
- 25 (a) $r < m$, always $r \leq n$ (b) $r = m, r < n$ (c) $r < m, r = n$ (d) $r = m = n$.
- 28 $\begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 0 & 4 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}; x_n = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}; \begin{bmatrix} 1 & 2 & 3 & 5 \\ 0 & 0 & 4 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$
Free $x_2 = 0$ gives $x_p = (-1, 0, 2)$ because the pivot columns contain I .
- 30 $\begin{bmatrix} 1 & 0 & 2 & 3 & 2 \\ 1 & 3 & 2 & 0 & 5 \\ 2 & 0 & 4 & 9 & 10 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 & 3 & 2 \\ 0 & 3 & 0 & -3 & 3 \\ 0 & 0 & 0 & 3 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 & 0 & -4 \\ 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}; \begin{bmatrix} -4 \\ 3 \\ 0 \\ 2 \end{bmatrix}; x_n = x_3 \begin{bmatrix} -2 \\ 0 \\ 1 \\ 0 \end{bmatrix}.$
- 36 If $Ax = b$ and $Cx = b$ have the same solutions, A and C have the same shape and the same nullspace (take $b = 0$). If $b =$ column 1 of A , $x = (1, 0, \dots, 0)$ solves $Ax = b$ so it solves $Cx = b$. Then A and C share column 1. Other columns too: $A = C$!

Problem Set 3.5, page 178

- 2 v_1, v_2, v_3 are independent (the -1 's are in different positions). All six vectors are on the plane $(1, 1, 1) \cdot v = 0$ so no four of these six vectors can be independent.
- 3 If $a = 0$ then column 1 = 0 ; if $d = 0$ then $b(\text{column 1}) - a(\text{column 2}) = 0$; if $f = 0$ then all columns end in zero (they are all in the xy plane, they must be dependent).
- 6 Columns 1, 2, 4 are independent. Also 1, 3, 4 and 2, 3, 4 and others (but not 1, 2, 3). Same column numbers (not same columns!) for A .
- 8 If $c_1(w_2 + w_3) + c_2(w_1 + w_3) + c_3(w_1 + w_2) = 0$ then $(c_2 + c_3)w_1 + (c_1 + c_3)w_2 + (c_1 + c_2)w_3 = 0$. Since the w 's are independent, $c_2 + c_3 = c_1 + c_3 = c_1 + c_2 = 0$. The only solution is $c_1 = c_2 = c_3 = 0$. Only this combination of v_1, v_2, v_3 gives 0 .
- 11 (a) Line in $\mathbf{R}^3$ (b) Plane in $\mathbf{R}^3$ (c) All of $\mathbf{R}^3$ (d) All of $\mathbf{R}^3$.
- 12 b is in the column space when $Ax = b$ has a solution; c is in the row space when $A^T y = c$ has a solution. *False*. The zero vector is always in the row space.
- 15 The n independent vectors span a space of dimension n . They are a *basis* for that space. If they are the columns of A then m is *not less* than n ($m \geq n$).
- 18 (a) The 6 vectors *might not* span $\mathbf{R}^4$ (b) The 6 vectors *are not* independent (c) Any four *might be* a basis.
- 20 One basis is $(2, 1, 0), (-3, 0, 1)$. A basis for the intersection with the xy plane is $(2, 1, 0)$. The normal vector $(1, -2, 3)$ is a basis for the line perpendicular to the plane.
- 22 (a) True (b) False because the basis vectors for $\mathbf{R}^6$ might not be in S .
- 25 Rank 2 if $c = 0$ and $d = 2$; rank 2 except when $c = d$ or $c = -d$.
- 28 $\begin{bmatrix} 1 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & -1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \end{bmatrix}; \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0 & -1 \\ -1 & 0 & 1 \end{bmatrix}.$
- 32 $y(0) = 0$ requires $A + B + C = 0$. One basis is $\cos x - \cos 2x$ and $\cos x - \cos 3x$.

- 34 $y_1(x), y_2(x), y_3(x)$ can be $x, 2x, 3x$ (dim 1) or $x, 2x, x^2$ (dim 2) or x, x^2, x^3 (dim 3).
- 37 The subspace of matrices that have $AS = SA$ has dimension *three*.
- 39 If the 5 by 5 matrix $[A \ b]$ is invertible, b is not a combination of the columns of A . If $[A \ b]$ is singular, and the 4 columns of A are independent, b is a combination of those columns. In this case $Ax = b$ has a solution.
- 41 $I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$. The six P 's are dependent.
- 42 The dimension of S is (a) zero when $x = 0$ (b) one when $x = (1, 1, 1, 1)$ (c) three when $x = (1, 1, -1, -1)$ because all rearrangements have $x_1 + \cdots + x_4 = 0$ (d) four when the x 's are not equal and don't add to zero. No x gives $\dim S = 2$.
- 43 The problem is to show that the u 's, v 's, w 's together are independent. We know the u 's and v 's together are a basis for V , and the u 's and w 's together are a basis for W . Suppose a combination of u 's, v 's, w 's gives 0 . **To be proved:** All coefficients = zero.
Key idea: The part x from the u 's and v 's is in V , so the part from the w 's is $-x$. This part is now in V and also in W . But if $-x$ is in $V \cap W$ it is a combination of u 's only. Now $x - x = 0$ uses only u 's and v 's (independent in V !) so all coefficients of u 's and v 's must be zero. Then $x = 0$ and the coefficients of the w 's are also zero.
- 44 The inputs to an m by n matrix fill $\mathbf{R}^n$. The outputs (column space!) have dimension r . The nullspace has $n - r$ special solutions. The formula becomes $r + (n - r) = n$.

Problem Set 3.6, page 190

- 1 (a) Row and column space dimensions = 5, nullspace dimension = 4, $\dim(N(A^T)) = 2$ sum = 16 = $m + n$ (b) Column space is $\mathbf{R}^3$; left nullspace contains only 0 .
- 4 (a) $\begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$ (b) Impossible: $r + (n - r)$ must be 3 (c) $[1 \ 1]$ (d) $\begin{bmatrix} -9 & -3 \\ 3 & 1 \end{bmatrix}$
 (e) *Impossible* Row space = column space requires $m = n$. Then $m - r = n - r$; nullspaces have the same dimension. Section 4.1 will prove $N(A)$ and $N(A^T)$ orthogonal to the row and column spaces respectively—here those are the same space.
- 6 A : dim 2, 2, 2, 1: Rows $(0, 3, 3, 3)$ and $(0, 1, 0, 1)$; columns $(3, 0, 1)$ and $(3, 0, 0)$; nullspace $(1, 0, 0, 0)$ and $(0, -1, 0, 1)$; $N(A^T)$ $(0, 1, 0)$. B : dim 1, 1, 0, 2 Row space (1) , column space $(1, 4, 5)$, nullspace: empty basis, $N(A^T)$ $(-4, 1, 0)$ and $(-5, 0, 1)$.
- 9 (a) Same row space and nullspace. So rank (dimension of row space) is the same (b) Same column space and left nullspace. Same rank (dimension of column space).
- 11 (a) No solution means that $r < m$. Always $r \leq n$. Can't compare m and n (b) Since $m - r > 0$, the left nullspace must contain a nonzero vector.
- 12 A neat choice is $\begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 1 \\ 2 & 4 & 0 \\ 1 & 0 & 1 \end{bmatrix}$; $r + (n - r) = n = 3$ does not match $2 + 2 = 4$. Only $v = 0$ is in both $N(A)$ and $C(A^T)$.
- 16 If $Av = 0$ and v is a row of A then $v \cdot v = 0$.

- 18 Row $3 - 2 \text{ row } 2 + \text{row } 1 = \text{zero row}$ so the vectors $c(1, -2, 1)$ are in the left nullspace. The same vectors happen to be in the nullspace (an accident for this matrix).
- 20 (a) Special solutions $(-1, 2, 0, 0)$ and $(-\frac{1}{4}, 0, -3, 1)$ are perpendicular to the rows of R (and then ER). (b) $A^T y = 0$ has 1 independent solution = last row of E^{-1} . ($E^{-1}A = R$ has a zero row, which is just the transpose of $A^T y = 0$).
- 21 (a) u and w (b) v and z (c) $\text{rank} < 2$ if u and w are dependent or if v and z are dependent (d) The rank of $uv^T + wz^T$ is 2.
- 24 $A^T y = d$ puts d in the row space of A ; unique solution if the left nullspace (nullspace of A^T) contains only $y = 0$.
- 26 The rows of $C = AB$ are combinations of the rows of B . So $\text{rank } C \leq \text{rank } B$. Also $\text{rank } C \leq \text{rank } A$, because the columns of C are combinations of the columns of A .
- 29 $a_{11} = 1, a_{12} = 0, a_{13} = 1, a_{22} = 0, a_{32} = 1, a_{31} = 0, a_{23} = 1, a_{33} = 0, a_{21} = 1$.
- 30 The subspaces for $A = uv^T$ are pairs of orthogonal lines (v and $v^\perp$, u and $u^\perp$). If B has those same four subspaces then $B = cA$ with $c \neq 0$.
- 31 (a) $AX = 0$ if each column of X is a multiple of $(1, 1, 1)$; $\dim(\text{nullspace}) = 3$. (b) If $AX = B$ then all columns of B add to zero; dimension of the B 's = 6. (c) $3 + 6 = \dim(M^{3 \times 3}) = 9$ entries in a 3 by 3 matrix.
- 32 The key is equal row spaces. First row of $A = \text{combination of the rows of } B$: only possible combination (notice I) is 1 (row 1 of B). Same for each row so $F = G$.

Problem Set 4.1, page 202

- 1 Both nullspace vectors are orthogonal to the row space vector in $\mathbf{R}^3$. The column space is perpendicular to the nullspace of A^T (two lines in $\mathbf{R}^2$ because $\text{rank} = 1$).
- 3 (a) $\begin{bmatrix} 1 & 2 & -3 \\ 2 & -3 & 1 \\ -3 & 5 & -2 \end{bmatrix}$ (b) Impossible, $\begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix}$ not orthogonal to $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ in $C(A)$ and $N(A^T)$ is impossible: not perpendicular (d) Need $A^2 = 0$; take $A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$ (e) $(1, 1, 1)$ in the nullspace (columns add to 0) and also row space; no such matrix.
- 6 Multiply the equations by $y_1, y_2, y_3 = 1, 1, -1$. Equations add to $0 = 1$ so no solution: $y = (1, 1, -1)$ is in the left nullspace. $Ax = b$ would need $0 = (y^T A)x = y^T b = 1$.
- 8 $x = x_r + x_n$, where x_r is in the row space and x_n is in the nullspace. Then $Ax_n = 0$ and $Ax = Ax_r + Ax_n = Ax_r$. All Ax are in $C(A)$.
- 9 Ax is always in the column space of A . If $A^T Ax = 0$ then Ax is also in the nullspace of A^T . So Ax is perpendicular to itself. Conclusion: $Ax = 0$ if $A^T Ax = 0$.
- 10 (a) With $A^T = A$, the column and row spaces are the same (b) x is in the nullspace and z is in the column space = row space: so these "eigenvectors" have $x^T z = 0$.
- 12 x splits into $x_r + x_n = (1, -1) + (1, 1) = (2, 0)$. Notice $N(A^T)$ is a plane $(1, 0) = (1, 1)/2 + (1, -1)/2 = x_r + x_n$.
- 13 $V^T W = \text{zero}$ makes each basis vector for V orthogonal to each basis vector for W . Then every v in V is orthogonal to every w in W (combinations of the basis vectors).

- 14 $Ax = B\hat{x}$ means that $[A \ B] \begin{bmatrix} x \\ -\hat{x} \end{bmatrix} = \mathbf{0}$. Three homogeneous equations in four unknowns always have a nonzero solution. Here $x = (3, 1)$ and $\hat{x} = (1, 0)$ and $Ax = B\hat{x} = (5, 6, 5)$ is in both column spaces. Two planes in $\mathbf{R}^3$ must share a line.
- 16 $A^T y = \mathbf{0}$ leads to $(Ax)^T y = x^T A^T y = 0$. Then $y \perp Ax$ and $N(A^T) \perp C(A)$.
- 18 $S^\perp$ is the nullspace of $A = \begin{bmatrix} 1 & 5 & 1 \\ 2 & 2 & 2 \end{bmatrix}$. Therefore $S^\perp$ is a *subspace* even if S is not.
- 21 For example $(-5, 0, 1, 1)$ and $(0, 1, -1, 0)$ span $S^\perp = \text{nullspace of } A = \begin{bmatrix} 1 & 2 & 2 & 3 \\ 1 & 3 & 3 & 2 \end{bmatrix}$.
- 23 x in $V^\perp$ is perpendicular to any vector in V . Since V contains all the vectors in S , x is also perpendicular to any vector in S . So every x in $V^\perp$ is also in $S^\perp$.
- 28 (a) $(1, -1, 0)$ is in both planes. Normal vectors are perpendicular, but planes still intersect! (b) Need *three* orthogonal vectors to span the whole orthogonal complement. (c) Lines can meet at the zero vector without being orthogonal.
- 30 When $AB = 0$, the column space of B is contained in the nullspace of A . Therefore the dimension of $C(B) \leq \text{dimension of } N(A)$. This means $\text{rank}(B) \leq 4 - \text{rank}(A)$.
- 31 $\text{null}(N')$ produces a basis for the *row space* of A (perpendicular to $N(A)$).
- 32 We need $r^T n = 0$ and $c^T \ell = 0$. All possible examples have the form acr^T with $a \neq 0$.
- 33 Both r 's orthogonal to both n 's, both c 's orthogonal to both ℓ 's, each pair independent. All A 's with these subspaces have the form $[c_1 \ c_2]M[r_1 \ r_2]^T$ for a 2 by 2 invertible M .

Problem Set 4.2, page 214

- 1 (a) $a^T b / a^T a = 5/3$; $p = 5a/3$; $e = (-2, 1, 1)/3$ (b) $a^T b / a^T a = -1$; $p = a$; $e = \mathbf{0}$.
- 3 $P_1 = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ and $P_1 b = \frac{1}{3} \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix}$. $P_2 = \frac{1}{11} \begin{bmatrix} 1 & 3 & 1 \\ 3 & 9 & 3 \\ 1 & 3 & 1 \end{bmatrix}$ and $P_2 b = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$.
- 6 $p_1 = (\frac{1}{9}, -\frac{2}{9}, -\frac{2}{9})$ and $p_2 = (\frac{4}{9}, \frac{4}{9}, -\frac{2}{9})$ and $p_3 = (\frac{4}{9}, -\frac{2}{9}, \frac{4}{9})$. So $p_1 + p_2 + p_3 = b$.
- 9 Since A is invertible, $P = A(A^T A)^{-1} A^T = A A^{-1} (A^T)^{-1} A^T = I$: project on all of $\mathbf{R}^2$.
- 11 (a) $p = A(A^T A)^{-1} A^T b = (2, 3, 0)$, $e = (0, 0, 4)$, $A^T e = \mathbf{0}$ (b) $p = (4, 4, 6)$, $e = \mathbf{0}$.
- 15 $2A$ has the same column space as A . $\hat{x}$ for $2A$ is *half* of $\hat{x}$ for A .
- 16 $\frac{1}{2}(1, 2, -1) + \frac{3}{2}(1, 0, 1) = (2, 1, 1)$. So b is in the plane. Projection shows $Pb = b$.
- 18 (a) $I - P$ is the projection matrix onto $(1, -1)$ in the perpendicular direction to $(1, 1)$ (b) $I - P$ projects onto the plane $x + y + z = 0$ perpendicular to $(1, 1, 1)$.
- 20 $e = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}$, $Q = \frac{ee^T}{e^T e} = \begin{bmatrix} 1/6 & -1/6 & -1/3 \\ -1/6 & 1/6 & 1/3 \\ -1/3 & 1/3 & 2/3 \end{bmatrix}$, $I - Q = \begin{bmatrix} 5/6 & 1/6 & 1/3 \\ 1/6 & 5/6 & -1/3 \\ 1/3 & -1/3 & 1/3 \end{bmatrix}$.
- 21 $(A(A^T A)^{-1} A^T)^2 = A(A^T A)^{-1} (A^T A) (A^T A)^{-1} A^T = A(A^T A)^{-1} A^T$. So $P^2 = P$. Pb is in the column space (where P projects). Then its projection $P(Pb)$ is Pb .

- 24 The nullspace of A^T is *orthogonal* to the column space $C(A)$. So if $A^T \mathbf{b} = \mathbf{0}$, the projection of $\mathbf{b}$ onto $C(A)$ should be $\mathbf{p} = \mathbf{0}$. Check $P\mathbf{b} = A(A^T A)^{-1} A^T \mathbf{b} = A(A^T A)^{-1} \mathbf{0}$.
- 28 $P^2 = P = P^T$ give $P^T P = P$. Then the $(2, 2)$ entry of P equals the $(2, 2)$ entry of $P^T P$ which is the length squared of column 2.
- 29 $A = B^T$ has independent columns, so $A^T A$ (which is BB^T) must be invertible.
- 30 (a) The column space is the line through $\mathbf{a} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ so $P_C = \frac{\mathbf{a}\mathbf{a}^T}{\mathbf{a}^T \mathbf{a}} = \frac{1}{25} \begin{bmatrix} 9 & 12 \\ 12 & 25 \end{bmatrix}$.
 (b) The row space is the line through $\mathbf{v} = (1, 2, 2)$ and $P_R = \mathbf{v}\mathbf{v}^T / \mathbf{v}^T \mathbf{v}$. Always $P_C A = A$ (columns of A project to themselves) and $AP_R = A$. Then $P_C AP_R = A$!
- 31 The error $\mathbf{e} = \mathbf{b} - \mathbf{p}$ must be perpendicular to all the $\mathbf{a}$'s.
- 32 Since $P_1 \mathbf{b}$ is in $C(A)$, $P_2(P_1 \mathbf{b})$ equals $P_1 \mathbf{b}$. So $P_2 P_1 = P_1 = \mathbf{a}\mathbf{a}^T / \mathbf{a}^T \mathbf{a}$ where $\mathbf{a} = (1, 2, 0)$.
- 33 If $P_1 P_2 = P_2 P_1$ then S is contained in T or T is contained in S .
- 34 BB^T is invertible as in Problem 29. Then $(A^T A)(BB^T) =$ product of r by r invertible matrices, so rank r . AB can't have rank $< r$, since A^T and B^T cannot increase the rank.
Conclusion: A (m by r of rank r) times B (r by n of rank r) produces AB of rank r .

Problem Set 4.3, page 226

- 1 $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 0 \\ 8 \\ 8 \\ 20 \end{bmatrix}$ give $A^T A = \begin{bmatrix} 4 & 8 \\ 8 & 26 \end{bmatrix}$ and $A^T \mathbf{b} = \begin{bmatrix} 36 \\ 112 \end{bmatrix}$.
- $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$ gives $\hat{\mathbf{x}} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$ and $\mathbf{p} = A\hat{\mathbf{x}} = \begin{bmatrix} 1 \\ 5 \\ 13 \\ 17 \end{bmatrix}$ and $\mathbf{e} = \mathbf{b} - \mathbf{p} = \begin{bmatrix} -1 \\ 3 \\ -5 \\ 3 \end{bmatrix}$
 $E = \|\mathbf{e}\|^2 = 44$
- 5 $E = (C-0)^2 + (C-8)^2 + (C-8)^2 + (C-20)^2$. $A^T = [1 \ 1 \ 1 \ 1]$ and $A^T A = [4]$.
 $A^T \mathbf{b} = [36]$ and $(A^T A)^{-1} A^T \mathbf{b} = 9 =$ best height C . Errors $\mathbf{e} = (-9, -1, -1, 11)$.
- 7 $A = [0 \ 1 \ 3 \ 4]^T$, $A^T A = [26]$ and $A^T \mathbf{b} = [112]$. Best $D = 112/26 = 56/13$.
- 8 $\hat{\mathbf{x}} = 56/13$, $\mathbf{p} = (56/13)(0, 1, 3, 4)$. $(C, D) = (9, 56/13)$ don't match $(C, D) = (1, 4)$.
 Columns of A were not perpendicular so we can't project separately to find C and D .
- 9 Parabola
 Project $\mathbf{b}$
 4D to 3D $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{bmatrix} \begin{bmatrix} C \\ D \\ E \end{bmatrix} = \begin{bmatrix} 0 \\ 8 \\ 8 \\ 20 \end{bmatrix}$. $A^T A \hat{\mathbf{x}} = \begin{bmatrix} 4 & 8 & 26 \\ 8 & 26 & 92 \\ 26 & 92 & 338 \end{bmatrix} \begin{bmatrix} C \\ D \\ E \end{bmatrix} = \begin{bmatrix} 36 \\ 112 \\ 400 \end{bmatrix}$.
- 11 (a) The best line $x = 1 + 4t$ gives the center point $\hat{\mathbf{b}} = 9$ when $\hat{t} = 2$.
 (b) The first equation $Cm + D \sum t_i = \sum b_i$ divided by m gives $C + D\hat{t} = \hat{\mathbf{b}}$.
- 13 $(A^T A)^{-1} A^T (\mathbf{b} - A\mathbf{x}) = \hat{\mathbf{x}} - \mathbf{x}$. When $\mathbf{e} = \mathbf{b} - A\mathbf{x}$ averages to $\mathbf{0}$, so does $\hat{\mathbf{x}} - \mathbf{x}$.
- 14 The matrix $(\hat{\mathbf{x}} - \mathbf{x})(\hat{\mathbf{x}} - \mathbf{x})^T$ is $(A^T A)^{-1} A^T (\mathbf{b} - A\mathbf{x})(\mathbf{b} - A\mathbf{x})^T A (A^T A)^{-1}$. When the average of $(\mathbf{b} - A\mathbf{x})(\mathbf{b} - A\mathbf{x})^T$ is $\sigma^2 I$, the average of $(\hat{\mathbf{x}} - \mathbf{x})(\hat{\mathbf{x}} - \mathbf{x})^T$ will be the output covariance matrix $(A^T A)^{-1} A^T \sigma^2 A (A^T A)^{-1}$ which simplifies to $\sigma^2 (A^T A)^{-1}$.

- 16 $\frac{1}{10}b_{10} + \frac{9}{10}\hat{x}_9 = \frac{1}{10}(b_1 + \cdots + b_{10})$. Knowing $\hat{x}_9$ avoids adding all b 's.
- 18 $p = A\hat{x} = (5, 13, 17)$ gives the heights of the closest line. The error is $b - p = (2, -6, 4)$. This error e has $Pe = Pb - Pp = p - p = 0$.
- 21 e is in $N(A^T)$; p is in $C(A)$; $\hat{x}$ is in $C(A^T)$; $N(A) = \{0\}$ = zero vector only.
- 23 The square of the distance between points on two lines is $E = (y - x)^2 + (3y - x)^2 + (1 + x)^2$. Derivatives $\frac{1}{2}\partial E/\partial x = 3x - 4y + 1 = 0$ and $\frac{1}{2}\partial E/\partial y = -4x + 10y = 0$. The solution is $x = -5/7, y = -2/7; E = 2/7$, and the minimum distance is $\sqrt{2/7}$.
- 25 3 points on a line: *Equal slopes* $(b_2 - b_1)/(t_2 - t_1) = (b_3 - b_2)/(t_3 - t_2)$. Linear algebra: Orthogonal to $(1, 1, 1)$ and (t_1, t_2, t_3) is $y = (t_2 - t_3, t_3 - t_1, t_1 - t_2)$ in the left nullspace. b is in the column space. Then $y^T b = 0$ is the same equal slopes condition written as $(b_2 - b_1)(t_3 - t_2) = (b_3 - b_2)(t_2 - t_1)$.
- 27 The shortest link connecting two lines in space is *perpendicular to those lines*.
- 28 Only 1 plane contains $0, a_1, a_2$ unless a_1, a_2 are *dependent*. Same test for $a_1, \dots, a_n$.

Problem Set 4.4, page 239

- 3 (a) $A^T A$ will be $16I$ (b) $A^T A$ will be diagonal with entries 1, 4, 9.
- 6 $Q_1 Q_2$ is orthogonal because $(Q_1 Q_2)^T Q_1 Q_2 = Q_2^T Q_1^T Q_1 Q_2 = Q_2^T Q_2 = I$.
- 8 If q_1 and q_2 are *orthonormal* vectors in $\mathbf{R}^5$ then $(q_1^T b)q_1 + (q_2^T b)q_2$ is closest to b .
- 11 (a) Two *orthonormal* vectors are $q_1 = \frac{1}{10}(1, 3, 4, 5, 7)$ and $q_2 = \frac{1}{10}(-7, 3, 4, -5, 1)$
 (b) Closest in the plane: *project* $Q Q^T(1, 0, 0, 0, 0) = (0.5, -0.18, -0.24, 0.4, 0)$.
- 13 The multiple to subtract is $\frac{a^T b}{a^T a}$. Then $B = b - \frac{a^T b}{a^T a}a = (4, 0) - 2 \cdot (1, 1) = (2, -2)$.
- 14 $\begin{bmatrix} 1 & 4 \\ 1 & 0 \end{bmatrix} = [q_1 \ q_2] \begin{bmatrix} \|a\| & q_1^T b \\ 0 & \|B\| \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} \sqrt{2} & 2\sqrt{2} \\ 0 & 2\sqrt{2} \end{bmatrix} = QR$.
- 15 (a) $q_1 = \frac{1}{3}(1, 2, -2), q_2 = \frac{1}{3}(2, 1, 2), q_3 = \frac{1}{3}(2, -2, -1)$ (b) The nullspace of A^T contains q_3 (c) $\hat{x} = (A^T A)^{-1} A^T(1, 2, 7) = (1, 2)$.
- 16 The projection $p = (a^T b / a^T a)a = 14a/49 = 2a/7$ is closest to b ; $q_1 = a/\|a\| = a/7$ is $(4, 5, 2, 2)/7$. $B = b - p = (-1, 4, -4, -4)/7$ has $\|B\| = 1$ so $q_2 = B$.
- 18 $A = a = (1, -1, 0, 0); B = b - p = (\frac{1}{2}, \frac{1}{2}, -1, 0); C = c - p_A - p_B = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -1)$. Notice the pattern in those orthogonal A, B, C . In $\mathbf{R}^5$, D would be $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -1)$.
- 20 (a) *True* (b) *True*. $Qx = x_1 q_1 + x_2 q_2$. $\|Qx\|^2 = x_1^2 + x_2^2$ because $q_1 \cdot q_2 = 0$.
- 21 The orthonormal vectors are $q_1 = (1, 1, 1, 1)/2$ and $q_2 = (-5, -1, 1, 5)/\sqrt{52}$. Then $b = (-4, -3, 3, 0)$ projects to $p = (-7, -3, -1, 3)/2$. And $b - p = (-1, -3, 7, -3)/2$ is orthogonal to both q_1 and q_2 .
- 22 $A = (1, 1, 2), B = (1, -1, 0), C = (-1, -1, 1)$. These are not yet unit vectors.
- 26 $(q_2^T C^*)q_2 = \frac{B^T c}{B^T B}B$ because $q_2 = \frac{B}{\|B\|}$ and the extra q_1 in C^* is orthogonal to q_2 .
- 28 There are mn multiplications in (11) and $\frac{1}{2}m^2n$ multiplications in each part of (12).

- 30 The wavelet matrix W has orthonormal columns. Notice $W^{-1} = W^T$ in Section 7.3.
- 32 $Q_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ reflects across x axis, $Q_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$ across plane $y + z = 0$.
- 33 Orthogonal and lower triangular $\Rightarrow \pm 1$ on the main diagonal and zeros elsewhere.

Problem Set 5.1, page 251

- 1 $\det(2A) = 8$; $\det(-A) = (-1)^4 \det A = \frac{1}{2}$; $\det(A^2) = \frac{1}{4}$; $\det(A^{-1}) = 2 = \det(A^T)^{-1}$.
- 5 $|J_5| = 1$, $|J_6| = -1$, $|J_7| = -1$. Determinants 1, 1, -1, -1 repeat so $|J_{101}| = 1$.
- 8 $Q^T Q = I \Rightarrow |Q|^2 = 1 \Rightarrow |Q| = \pm 1$; Q^n stays orthogonal so $\det$ can't blow up.
- 10 If the entries in every row add to zero, then $(1, 1, \dots, 1)$ is in the nullspace: singular A has $\det = 0$. (The columns add to the zero column so they are linearly dependent.)
If every row adds to one, then rows of $A - I$ add to zero (not necessarily $\det A = 1$).
- 11 $CD = -DC \Rightarrow \det CD = (-1)^n \det DC$ and *not* $-\det DC$. If n is even we can have an invertible CD .
- 14 $\det(A) = 36$ and the 4 by 4 second difference matrix has $\det = 5$.
- 15 The first determinant is 0, the second is $1 - 2t^2 + t^4 = (1 - t^2)^2$.
- 17 Any 3 by 3 skew-symmetric K has $\det(K^T) = \det(-K) = (-1)^3 \det(K)$. This is $-\det(K)$. But always $\det(K^T) = \det(K)$, so we must have $\det(K) = 0$ for 3 by 3.
- 21 Rules 5 and 3 give Rule 2. (Since Rules 4 and 3 give 5, they also give Rule 2.)
- 23 $\det(A) = 10$, $A^2 = \begin{bmatrix} 18 & 7 \\ 14 & 11 \end{bmatrix}$, $\det(A^2) = 100$, $A^{-1} = \frac{1}{10} \begin{bmatrix} 3 & -1 \\ -2 & 4 \end{bmatrix}$ has $\det \frac{1}{10}$.
 $\det(A - \lambda I) = \lambda^2 - 7\lambda + 10 = 0$ when $\lambda = 2$ or $\lambda = 5$; those are eigenvalues.
- 27 $\det A = abc$, $\det B = -abcd$, $\det C = a(b - a)(c - b)$ by doing elimination.
- 32 Typical determinants of $\text{rand}(n)$ are $10^6, 10^{25}, 10^{79}, 10^{218}$ for $n = 50, 100, 200, 400$.
 $\text{randn}(n)$ with normal distribution gives $10^{31}, 10^{78}, 10^{186}$, Inf which means $\geq 2^{1024}$.
MATLAB allows $1.9999999999999999 \times 2^{1023} \approx 1.8 \times 10^{308}$ but one more 9 gives Inf!

Problem Set 5.2, page 263

- 2 $\det A = -2$, independent; $\det B = 0$, dependent; $\det C = -1$, independent.
- 4 $a_{11}a_{23}a_{32}a_{44}$ gives -1 , because $2 \leftrightarrow 3$, $a_{14}a_{23}a_{32}a_{41}$ gives $+1$, $\det A = 1 - 1 = 0$;
 $\det B = 2 \cdot 4 \cdot 4 \cdot 2 - 1 \cdot 4 \cdot 4 \cdot 1 = 64 - 16 = 48$.
- 6 (a) If $a_{11} = a_{22} = a_{33} = 0$ then 4 terms are sure zeros (b) 15 terms must be zero.
- 8 Some term $a_{1\alpha}a_{2\beta} \cdots a_{n\omega}$ in the big formula is not zero! Move rows $1, 2, \dots, n$ into rows $\alpha, \beta, \dots, \omega$. Then these nonzero a 's will be on the main diagonal.
- 9 To get $+1$ for the even permutations the matrix needs an *even* number of -1 's. For the odd P 's the matrix needs an *odd* number of -1 's. So six 1 's and $\det = 6$ are impossible five 1 's and one -1 will give $AC = (ad - bc)I = (\det A)I$ $\max(\det) = 4$.

- 11 $C = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. $D = \begin{bmatrix} 0 & 42 & -35 \\ 0 & -21 & 14 \\ -3 & 6 & -3 \end{bmatrix}$. $\det B = 1(0) + 2(42) + 3(-35) = -21$.
Puzzle: $\det D = 441 = (-21)^2$. Why?
- 12 $C = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}$ and $AC^T = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$. Therefore $A^{-1} = \frac{1}{4}C^T = C^T/\det A$.
- 13 (a) $C_1 = 0$, $C_2 = -1$, $C_3 = 0$, $C_4 = 1$ (b) $C_n = -C_{n-2}$ by cofactors of row 1 then cofactors of column 1. Therefore $C_{10} = -C_8 = C_6 = -C_4 = C_2 = -1$.
- 15 The 1, 1 cofactor of the n by n matrix is E_{n-1} . The 1, 2 cofactor has a single 1 in its first column, with cofactor E_{n-2} : sign gives $-E_{n-2}$. So $E_n = E_{n-1} - E_{n-2}$. Then E_1 to E_6 is 1, 0, -1, -1, 0, 1 and this cycle of six will repeat: $E_{100} = E_4 = -1$.
- 16 The 1, 1 cofactor of the n by n matrix is F_{n-1} . The 1, 2 cofactor has a 1 in column 1, with cofactor F_{n-2} . Multiply by $(-1)^{1+2}$ and also (-1) from the 1, 2 entry to find $F_n = F_{n-1} + F_{n-2}$ (so these determinants are Fibonacci numbers).
- 19 Since x , x^2 , x^3 are all in the same row, they are never multiplied in $\det V_4$. The determinant is zero at $x = a$ or b or c , so $\det V$ has factors $(x-a)(x-b)(x-c)$. Multiply by the cofactor V_3 . The Vandermonde matrix $V_{ij} = (x_i)^{j-1}$ is for fitting a polynomial $p(x) = b$ at the points x_i . It has $\det V = \text{product of all } x_k - x_m \text{ for } k > m$.
- 20 $G_2 = -1$, $G_3 = 2$, $G_4 = -3$, and $G_n = (-1)^{n-1}(n-1) = (\text{product of the } \lambda\text{'s})$.
- 24 (a) All L 's have $\det = 1$; $\det U_k = \det A_k = 2, 6, -6$ (b) Pivots 5, 6/5, 7/6.
- 25 Problem 23 gives $\det \begin{bmatrix} I & 0 \\ -CA^{-1} & I \end{bmatrix} = 1$ and $\det \begin{bmatrix} A & B \\ C & D \end{bmatrix} = |A| \text{ times } |D - CA^{-1}B|$ which is $|AD - ACA^{-1}B|$. If $AC = CA$ this is $|AD - CAA^{-1}B| = \det(AD - CB)$.
- 27 (a) $\det A = a_{11}C_{11} + \cdots + a_{1n}C_{1n}$. Derivative with respect to $a_{11} = \text{cofactor } C_{11}$.
- 29 There are five nonzero products, all 1's with a plus or minus sign. Here are the (row, column) numbers and the signs: $+(1, 1)(2, 2)(3, 3)(4, 4) + (1, 2)(2, 1)(3, 4)(4, 3) - (1, 2)(2, 1)(3, 3)(4, 4) - (1, 1)(2, 2)(3, 4)(4, 3) - (1, 1)(2, 3)(3, 2)(4, 4)$. Total -1.
- 32 The problem is to show that $F_{2n+2} = 3F_{2n} - F_{2n-2}$. Keep using Fibonacci's rule:
 $F_{2n+2} = F_{2n+1} + F_{2n} = F_{2n} + F_{2n-1} + F_{2n} = 2F_{2n} + (F_{2n} - F_{2n-2}) = 3F_{2n} - F_{2n-2}$.
- 33 The difference from 20 to 19 multiplies its 3 by 3 cofactor = 1: then $\det$ drops by 1.
- 34 (a) The last three rows must be dependent (b) In each of the 120 terms: Choices from the last 3 rows must use 3 columns; at least one of those choices will be zero.

Problem Set 5.3, page 278

- 2 (a) $y = \begin{vmatrix} a & 1 \\ c & 0 \end{vmatrix} / \begin{vmatrix} a & b \\ c & d \end{vmatrix} = c/(ad - bc)$ (b) $y = \det B_2 / \det A = (fg - id)/D$.
- 3 (a) $x_1 = 3/0$ and $x_2 = -2/0$: no solution (b) $x_1 = x_2 = 0/0$: undetermined.
- 4 (a) $x_1 = \det([b \ a_2 \ a_3]) / \det A$, if $\det A \neq 0$ (b) The determinant is linear in its first column so $x_1|a_1 \ a_2 \ a_3| + x_2|a_2 \ a_2 \ a_3| + x_3|a_3 \ a_2 \ a_3|$. The last two determinants are zero because of repeated columns, leaving $x_1|a_1 \ a_2 \ a_3|$ which is $x_1 \det A$.

- 6 (a) $\begin{bmatrix} 1 & -\frac{2}{3} & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & -\frac{7}{3} & 1 \end{bmatrix}$ (b) $\frac{1}{4} \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}$. An invertible symmetric matrix has a symmetric inverse.
- 8 $C = \begin{bmatrix} 6 & -3 & 0 \\ 3 & 1 & -1 \\ -6 & 2 & 1 \end{bmatrix}$ and $AC^T = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$. This is $(\det A)I$ and $\det A = 3$. The 1, 3 cofactor of A is 0. Multiplying by 4 or 100: no change.
- 9 If we know the cofactors and $\det A = 1$, then $C^T = A^{-1}$ and also $\det A^{-1} = 1$. Now A is the inverse of C^T , so A can be found from the cofactor matrix for C .
- 11 The cofactors of A are integers. Division by $\det A = \pm 1$ gives integer entries in A^{-1} .
- 15 For $n = 5$, C contains 25 cofactors and each 4 by 4 cofactor has 24 terms. Each term needs 3 multiplications: total 1800 multiplications vs. 125 for Gauss-Jordan.
- 17 Volume = $\begin{vmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{vmatrix} = 20$. Area of faces length of cross product = $\begin{vmatrix} i & j & k \\ 3 & 1 & 1 \\ 1 & 3 & 1 \end{vmatrix} = \begin{matrix} -2i - 2j + 8k \\ \text{length} = 6\sqrt{2} \end{matrix}$
- 18 (a) Area $\frac{1}{2} \begin{vmatrix} 2 & 1 & 1 \\ 3 & 4 & 1 \\ 0 & 5 & 1 \end{vmatrix} = 5$ (b) $5 + \text{new triangle area } \frac{1}{2} \begin{vmatrix} 2 & 1 & 1 \\ 0 & 5 & 1 \\ -1 & 0 & 1 \end{vmatrix} = 5 + 7 = 12$.
- 21 The maximum volume is $L_1 L_2 L_3 L_4$ reached when the edges are orthogonal in $\mathbf{R}^4$. With entries 1 and -1 all lengths are $\sqrt{4} = 2$. The maximum determinant is $2^4 = 16$, achieved in Problem 20. For a 3 by 3 matrix, $\det A = (\sqrt{3})^3$ can't be achieved.
- 23 $A^T A = \begin{bmatrix} a^T \\ b^T \\ c^T \end{bmatrix} \begin{bmatrix} a & b & c \end{bmatrix} = \begin{bmatrix} a^T a & 0 & 0 \\ 0 & b^T b & 0 \\ 0 & 0 & c^T c \end{bmatrix}$ has $\begin{matrix} \det A^T A & = & (\|a\| \|b\| \|c\|)^2 \\ \det A & = & \pm \|a\| \|b\| \|c\| \end{matrix}$
- 25 The n -dimensional cube has 2^n corners, $n2^{n-1}$ edges and $2n(n-1)$ -dimensional faces. Coefficients from $(2+x)^n$ in Worked Example 2.4A. Cube from $2I$ has volume 2^n .
- 26 The pyramid has volume $\frac{1}{6}$. The 4-dimensional pyramid has volume $\frac{1}{24}$ (and $\frac{1}{n!}$ in $\mathbf{R}^n$).
- 31 Base area 10, height 2, volume 20.
- 35 $S = (2, 1, -1)$, area $\|PQ \times PS\| = \|(-2, -2, -1)\| = 3$. The other four corners can be $(0, 0, 0)$, $(0, 0, 2)$, $(1, 2, 2)$, $(1, 1, 0)$. The volume of the tilted box is $|\det| = 1$.
- 39 $AC^T = (\det A)I$ gives $(\det A)(\det C) = (\det A)^n$. Then $\det A = (\det C)^{1/3}$ with $n = 4$. With $\det A^{-1}$ is $1/\det A$, construct A^{-1} using the cofactors. *Invert to find A .*

Problem Set 6.1, page 293

- 1 The eigenvalues are 1 and 0.5 for A , 1 and 0.25 for A^2 , 1 and 0 for A^∞ . Exchanging the rows of A changes the eigenvalues to 1 and -0.5 (the trace is now $0.2 + 0.3$). Singular matrices stay singular during elimination, so $\lambda = 0$ does not change.
- 3 A has $\lambda_1 = 2$ and $\lambda_2 = -1$ (check trace and determinant) with $x_1 = (1, 1)$ and $x_2 = (2, -1)$. A^{-1} has the same eigenvectors, with eigenvalues $1/\lambda = \frac{1}{2}$ and -1 .
- 6 A and B have $\lambda_1 = 1$ and $\lambda_2 = 1$. AB and BA have $\lambda = 2 \pm \sqrt{3}$. Eigenvalues of AB are not equal to eigenvalues of A times eigenvalues of B . Eigenvalues of AB and BA are equal (this is proved in section 6.6, Problems 18-19).

- 8 (a) Multiply Ax to see λx which reveals λ (b) Solve $(A - \lambda I)x = 0$ to find x .
- 10 A has $\lambda_1 = 1$ and $\lambda_2 = .4$ with $x_1 = (1, 2)$ and $x_2 = (1, -1)$. A^∞ has $\lambda_1 = 1$ and $\lambda_2 = 0$ (same eigenvectors). A^{100} has $\lambda_1 = 1$ and $\lambda_2 = (.4)^{100}$ which is near zero. So A^{100} is very near A^∞ : same eigenvectors and close eigenvalues.
- 11 Columns of $A - \lambda_1 I$ are in the nullspace of $A - \lambda_2 I$ because $M = (A - \lambda_2 I)(A - \lambda_1 I) = \text{zero matrix}$ [this is the *Cayley-Hamilton Theorem* in Problem 6.2.32]. Notice that M has zero eigenvalues $(\lambda_1 - \lambda_2)(\lambda_1 - \lambda_1) = 0$ and $(\lambda_2 - \lambda_2)(\lambda_2 - \lambda_1) = 0$.
- 13 (a) $Pu = (uu^T)u = u(u^T u) = u$ so $\lambda = 1$ (b) $Pv = (uu^T)v = u(u^T v) = 0$
(c) $x_1 = (-1, 1, 0, 0)$, $x_2 = (-3, 0, 1, 0)$, $x_3 = (-5, 0, 0, 1)$ all have $Px = 0x = 0$.
- 15 The other two eigenvalues are $\lambda = \frac{1}{2}(-1 \pm i\sqrt{3})$; the three eigenvalues are $1, 1, -1$.
- 16 Set $\lambda = 0$ in $\det(A - \lambda I) = (\lambda_1 - \lambda) \cdots (\lambda_n - \lambda)$ to find $\det A = (\lambda_1)(\lambda_2) \cdots (\lambda_n)$.
- 17 $\lambda_1 = \frac{1}{2}(a + d + \sqrt{(a-d)^2 + 4bc})$ and $\lambda_2 = \frac{1}{2}(a + d - \sqrt{(a-d)^2 + 4bc})$ add to $a + d$.
If A has $\lambda_1 = 3$ and $\lambda_2 = 4$ then $\det(A - \lambda I) = (\lambda - 3)(\lambda - 4) = \lambda^2 - 7\lambda + 12$.
- 19 (a) $\text{rank} = 2$ (b) $\det(B^T B) = 0$ (d) eigenvalues of $(B^2 + I)^{-1}$ are $1, \frac{1}{2}, \frac{1}{5}$.
- 20 Last rows are $-28, 11$ (check trace and det) and $6, -11, 6$ (to match $\det(C - \lambda I)$).
- 22 $\lambda = 1$ (for Markov), 0 (for singular), $-\frac{1}{2}$ (so sum of eigenvalues = trace = $\frac{1}{2}$).
- 23 $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix}$. Always A^2 is the zero matrix if $\lambda = 0$ and 0 , by the Cayley-Hamilton Theorem in Problem 6.2.32.
- 28 B has $\lambda = -1, -1, -1, 3$ and C has $\lambda = 1, 1, 1, -3$. Both have $\det = -3$.
- 32 (a) u is a basis for the nullspace, v and w give a basis for the column space
(b) $x = (0, \frac{1}{3}, \frac{1}{5})$ is a particular solution. Add any cu from the nullspace
(c) If $Ax = u$ had a solution, u would be in the column space: wrong dimension 3.
- 34 $\det(P - \lambda I) = 0$ gives the equation $\lambda^4 = 1$. This reflects the fact that $P^4 = I$. The solutions of $\lambda^4 = 1$ are $\lambda = 1, i, -1, -i$. The real eigenvector $x_1 = (1, 1, 1, 1)$ is not changed by the permutation P . Three more eigenvectors are (i, i^2, i^3, i^4) and $(1, -1, 1, -1)$ and $(-i, (-i)^2, (-i)^3, (-i)^4)$.
- 36 $\lambda_1 = e^{2\pi i/3}$ and $\lambda_2 = e^{-2\pi i/3}$ give $\det \lambda_1 \lambda_2 = 1$ and trace $\lambda_1 + \lambda_2 = -1$.
 $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ with $\theta = \frac{2\pi}{3}$ has this trace and det. So does every $M^{-1}AM$!

Problem Set 6.2, page 307

$$^1 \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}; \begin{bmatrix} 1 & 1 \\ 3 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} \frac{3}{4} & -\frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} \end{bmatrix}.$$

- 3 If $A = S\Lambda S^{-1}$ then the eigenvalue matrix for $A + 2I$ is $\Lambda + 2I$ and the eigenvector matrix is still S . $A + 2I = S(\Lambda + 2I)S^{-1} = S\Lambda S^{-1} + S(2I)S^{-1} = A + 2I$.
- 4 (a) False: don't know λ 's (b) True (c) True (d) False: need eigenvectors of S
- 6 The columns of S are nonzero multiples of $(2, 1)$ and $(0, 1)$: either order. Same for A^{-1} .

- 8 $A = S\Lambda S^{-1} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{bmatrix}$. $S\Lambda^k S^{-1} = \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1^k & 0 \\ 0 & \lambda_2^k \end{bmatrix} \begin{bmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2nd \text{ component is } F_k \\ (\lambda_1^k - \lambda_2^k)/(\lambda_1 - \lambda_2) \end{bmatrix}$.
- 9 (a) $A = \begin{bmatrix} .5 & .5 \\ 1 & 0 \end{bmatrix}$ has $\lambda_1 = 1$, $\lambda_2 = -\frac{1}{2}$ with $x_1 = (1, 1)$, $x_2 = (1, -2)$
- (b) $A^n = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 1^n & 0 \\ 0 & (-.5)^n \end{bmatrix} \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & -\frac{1}{3} \end{bmatrix} \rightarrow A^\infty = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix}$
- 12 (a) False: don't know λ (b) True: an eigenvector is missing (c) True.
- 13 $A = \begin{bmatrix} 8 & 3 \\ -3 & 2 \end{bmatrix}$ (or other), $A = \begin{bmatrix} 9 & 4 \\ -4 & 1 \end{bmatrix}$, $A = \begin{bmatrix} 10 & 5 \\ -5 & 0 \end{bmatrix}$; only eigenvectors are $x = (c, -c)$.
- 15 $A^k = S\Lambda^k S^{-1}$ approaches zero if and only if every $|\lambda| < 1$; $A_1^k \rightarrow A_1^\infty$, $A_2^k \rightarrow 0$.
- 17 $\Lambda = \begin{bmatrix} .9 & 0 \\ 0 & .3 \end{bmatrix}$, $S = \begin{bmatrix} 3 & -3 \\ 1 & 1 \end{bmatrix}$; $A_2^{10} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = (.9)^{10} \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, $A_2^{10} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = (.3)^{10} \begin{bmatrix} 3 \\ -1 \end{bmatrix}$, $A_2^{10} \begin{bmatrix} 6 \\ 0 \end{bmatrix} = (.9)^{10} \begin{bmatrix} 3 \\ 1 \end{bmatrix} + (.3)^{10} \begin{bmatrix} 3 \\ -1 \end{bmatrix}$ because $\begin{bmatrix} 6 \\ 0 \end{bmatrix}$ is the sum of $\begin{bmatrix} 3 \\ 1 \end{bmatrix} + \begin{bmatrix} 3 \\ -1 \end{bmatrix}$.
- 19 $B^k = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & 4 \end{bmatrix}^k \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 5^k & 5^k - 4^k \\ 0 & 4^k \end{bmatrix}$.
- 21 trace $ST = (aq + bs) + (cr + dt)$ is equal to $(qa + rc) + (sb + td) = \text{trace } TS$.
Diagonalizable case: the trace of $S\Lambda S^{-1} = \text{trace of } (\Lambda S^{-1})S = \Lambda$: *sum of the λ 's*.
- 24 The A 's form a subspace since cA and $A_1 + A_2$ all have the same S . When $S = I$ the A 's with those eigenvectors give the subspace of diagonal matrices. Dimension 4.
- 26 Two problems: The nullspace and column space can overlap, so x could be in both. There may not be r independent eigenvectors in the column space.
- 27 $R = S\sqrt{\Lambda}S^{-1} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ has $R^2 = A$. $\sqrt{B}$ needs $\lambda = \sqrt{9}$ and $\sqrt{-1}$, trace is not real.
Note that $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ can have $\sqrt{-1} = i$ and $-i$, trace 0, real square root $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$.
- 28 $A^T = A$ gives $x^T A B x = (Ax)^T (Bx) \leq \|Ax\| \|Bx\|$ by the Schwarz inequality.
 $B^T = -B$ gives $-x^T B A x = (Bx)^T (Ax) \leq \|Ax\| \|Bx\|$. Add to get Heisenberg's Uncertainty Principle when $AB - BA = I$. Position-momentum, also time-energy.
- 32 If $A = S\Lambda S^{-1}$ then $(A - \lambda_1 I) \cdots (A - \lambda_n I)$ equals $S(\Lambda - \lambda_1 I) \cdots (\Lambda - \lambda_n I)S^{-1}$.
The factor $\Lambda - \lambda_j I$ is zero in row j . *The product is zero in all rows = zero matrix.*
- 33 $\lambda = 2, -1, 0$ are in Λ and the eigenvectors are in S (below). $A^k = S\Lambda^k S^{-1}$ is

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & -1 & 1 \\ 1 & -1 & -1 \end{bmatrix} \Lambda^k \frac{1}{6} \begin{bmatrix} 2 & 1 & 1 \\ 2 & -2 & -2 \\ 0 & 3 & -3 \end{bmatrix} = \frac{2^k}{6} \begin{bmatrix} 4 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{bmatrix} + \frac{(-1)^k}{3} \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

Check $k = 4$. The $(2, 2)$ entry of A^4 is $2^4/6 + (-1)^4/3 = 18/6 = 3$. The 4-step paths that begin and end at node 2 are 2 to 1 to 1 to 1 to 2, 2 to 1 to 2 to 1 to 2, and 2 to 1 to 3 to 1 to 2. Much harder to find the eleven 4-step paths that start and end at node 1.

35 B has $\lambda = i$ and $-i$, so B^4 has $\lambda^4 = 1$ and 1 and $B^4 = I$. C has $\lambda = (1 \pm \sqrt{3}i)/2$. This is $\exp(\pm\pi i/3)$ so $\lambda^3 = -1$ and -1 . Then $C^3 = -I$ and $C^{1024} = -C$.

37 Columns of S times rows of ΛS^{-1} will give r rank-1 matrices ($r = \text{rank of } A$).

Problem Set 6.3, page 325

- 1** $u_1 = e^{4t} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $u_2 = e^t \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. If $u(0) = (5, -2)$, then $u(t) = 3e^{4t} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2e^t \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.
- 4** $d(v+w)/dt = (w-v) + (v-w) = 0$, so the total $v+w$ is constant. $A = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$ has $\lambda_1 = 0$ with $x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $x_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$; $v(1) = 20 + 10e^{-2}$ $v(\infty) = 20$
 $\lambda_2 = -2$ $w(1) = 20 - 10e^{-2}$ $w(\infty) = 20$
- 8** $\begin{bmatrix} 6 & -2 \\ 2 & 1 \end{bmatrix}$ has $\lambda_1 = 5$, $x_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\lambda_2 = 2$, $x_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$; rabbits $r(t) = 20e^{5t} + 10e^{2t}$, $w(t) = 10e^{5t} + 20e^{2t}$. The ratio of rabbits to wolves approaches $20/10$; e^{5t} dominates.
- 12** $A = \begin{bmatrix} 0 & 1 \\ -9 & 6 \end{bmatrix}$ has trace 6, det 9, $\lambda = 3$ and 3 with *one* independent eigenvector $(1, 3)$.
- 14** When A is skew-symmetric, $\|u(t)\| = \|e^{At}u(0)\|$ is $\|u(0)\|$. So e^{At} is *orthogonal*.
- 15** $u_p = 4$ and $u(t) = ce^t + 4$; $u_p = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$ and $u(t) = c_1e^t \begin{bmatrix} 1 \\ t \end{bmatrix} + c_2e^t \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 4 \\ 2 \end{bmatrix}$.
- 16** Substituting $u = e^{ct}v$ gives $ce^{ct}v = Ae^{ct}v - e^{ct}b$ or $(A - cI)v = b$ or $v = (A - cI)^{-1}b = \text{particular solution}$. If c is an eigenvalue then $A - cI$ is not invertible.
- 20** The solution at time $t + T$ is also $e^{A(t+T)}u(0)$. Thus e^{At} times e^{AT} equals $e^{A(t+T)}$.
- 21** $\begin{bmatrix} 1 & 4 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 0 & -1 \end{bmatrix}$; $\begin{bmatrix} 1 & 4 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} e^t & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} e^t & 4e^t - 4 \\ 0 & 1 \end{bmatrix}$.
- 22** $A^2 = A$ gives $e^{At} = I + At + \frac{1}{2}At^2 + \cdots = I + (e^t - 1)A = \begin{bmatrix} e^t & e^t - 1 \\ 0 & 1 \end{bmatrix}$.
- 24** $A = \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix}$. Then $e^{At} = \begin{bmatrix} e^t & \frac{1}{2}(e^{3t} - e^t) \\ 0 & e^{3t} \end{bmatrix}$.
- 26** (a) The inverse of e^{At} is e^{-At} (b) If $Ax = \lambda x$ then $e^{At}x = e^{\lambda t}x$ and $e^{\lambda t} \neq 0$.
- 27** $(x, y) = (e^{4t}, e^{-4t})$ is a growing solution. The correct matrix for the exchanged $u = (y, x)$ is $\begin{bmatrix} 2 & -2 \\ -4 & 0 \end{bmatrix}$. It *does* have the same eigenvalues as the original matrix.
- 28** Centering produces $U_{n+1} = \begin{bmatrix} 1 & \Delta t \\ -\Delta t & 1 - (\Delta t)^2 \end{bmatrix} U_n$. At $\Delta t = 1$, $\lambda = e^{i\pi/3}$ and $e^{-i\pi/3}$ both have $\lambda^6 = 1$ so $A^6 = I$. $U_6 = A^6 U_0$ comes exactly back to U_0 .
- 29** First A has $\lambda = \pm i$ and $A^4 = I$ $A^n = (-1)^n \begin{bmatrix} 1 - 2n & -2n \\ 2n & 2n + 1 \end{bmatrix}$ Linear growth.
 Second A has $\lambda = -1, -1$ and

30 With $a = \Delta t/2$ the trapezoidal step is $U_{n+1} = \frac{1}{1+a^2} \begin{bmatrix} 1-a^2 & 2a \\ -2a & 1-a^2 \end{bmatrix} U_n$.

Orthonormal columns $\Rightarrow$ orthogonal matrix $\Rightarrow \|U_{n+1}\| = \|U_n\|$

- 31 (a) $(\cos A)x = (\cos \lambda)x$ (b) $\lambda(A) = 2\pi$ and 0 so $\cos \lambda = 1, 1$ and $\cos A = I$
 (c) $u(t) = 3(\cos 2\pi t)(1, 1) + 1(\cos 0t)(1, -1)$ [$u' = Au$ has **exp**, $u'' = Au$ has **cos**]

Problem Set 6.4, page 337

- 3 $\lambda = 0, 4, -2$; unit vectors $\pm(0, 1, -1)/\sqrt{2}$ and $\pm(2, 1, 1)/\sqrt{6}$ and $\pm(1, -1, -1)/\sqrt{3}$.

5 $Q = \frac{1}{3} \begin{bmatrix} 2 & 1 & 2 \\ 2 & -2 & -1 \\ -1 & -2 & 2 \end{bmatrix}$. The columns of Q are unit eigenvectors of A
 Each unit eigenvector could be multiplied by -1

- 8 If $A^3 = 0$ then all $\lambda^3 = 0$ so all $\lambda = 0$ as in $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. If A is symmetric then $A^3 = Q\Lambda^3Q^T = 0$ gives $\Lambda = 0$. The only symmetric A is $Q0Q^T =$ zero matrix.

- 10 If x is not real then $\lambda = x^T Ax / x^T x$ is *not* always real. Can't assume real eigenvectors!

11 $\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} = 2 \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} + 4 \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$; $\begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix} = 0 \begin{bmatrix} .64 & -.48 \\ -.48 & .36 \end{bmatrix} + 25 \begin{bmatrix} .36 & .48 \\ .48 & .64 \end{bmatrix}$

- 14 M is skew-symmetric and orthogonal; λ 's must be $i, i, -i, -i$ to have trace zero.

- 16 (a) If $Az = \lambda y$ and $A^T y = \lambda z$ then $B[y; -z] = [-Az; A^T y] = -\lambda[y; -z]$. So $-\lambda$ is also an eigenvalue of B . (b) $A^T Az = A^T(\lambda y) = \lambda^2 z$. (c) $\lambda = -1, -1, 1, 1$; $x_1 = (1, 0, -1, 0)$, $x_2 = (0, 1, 0, -1)$, $x_3 = (1, 0, 1, 0)$, $x_4 = (0, 1, 0, 1)$.

19 A has $S = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$; B has $S = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 2d \end{bmatrix}$. Perpendicular for A
 Not perpendicular for B
 since $B^T \neq B$

21 (a) False. $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ (b) True from $A^T = Q\Lambda Q^T$ (c) True from $A^{-1} = Q\Lambda^{-1}Q^T$ (d) False!

- 22 A and A^T have the same λ 's but the *order* of the x 's can change. $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ has $\lambda_1 = i$ and $\lambda_2 = -i$ with $x_1 = (1, i)$ first for A but $x_1 = (1, -i)$ first for A^T .

- 23 A is invertible, orthogonal, permutation, diagonalizable, Markov; B is projection, diagonalizable, Markov. A allows $QR, S\Lambda S^{-1}, Q\Lambda Q^T$; B allows $S\Lambda S^{-1}$ and $Q\Lambda Q^T$.

- 24 Symmetry gives $Q\Lambda Q^T$ if $b = 1$; repeated λ and no S if $b = -1$; singular if $b = 0$.

25 Orthogonal and symmetric requires $|\lambda| = 1$ and λ real, so $\lambda = \pm 1$. Then $A = \pm I$ or $A = Q\Lambda Q^T = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}$.

- 27 The roots of $\lambda^2 + b\lambda + c = 0$ differ by $\sqrt{b^2 - 4c}$. For $\det(A + tB - \lambda I)$ we have $b = -3 - 8t$ and $c = 2 + 16t - t^2$. The minimum of $b^2 - 4c$ is $1/17$ at $t = 2/17$. Then $\lambda_2 - \lambda_1 = 1/\sqrt{17}$.

- 29 (a) $A = Q\Lambda\bar{Q}^T$ times $\bar{A}^T = Q\bar{\Lambda}^T\bar{Q}^T$ equals $\bar{A}^T$ times A because $\Lambda\bar{\Lambda}^T = \bar{\Lambda}^T\Lambda$ (diagonal!) (b) step 2: The 1, 1 entries of $\bar{T}^T T$ and $T\bar{T}^T$ are $|a|^2$ and $|a|^2 + |b|^2$. This makes $b = 0$ and $T = \Lambda$.
- 30 a_{11} is $[q_{11} \dots q_{1n}] [\lambda_1 \bar{q}_{11} \dots \lambda_n \bar{q}_{1n}]^T \leq \lambda_{\max} (|q_{11}|^2 + \dots + |q_{1n}|^2) = \lambda_{\max}$.
- 31 (a) $x^T(Ax) = (Ax)^T x = x^T A^T x = -x^T Ax$. (b) $\bar{z}^T Az$ is pure imaginary, its real part is $x^T Ax + y^T Ay = 0 + 0$ (c) $\det A = \lambda_1 \dots \lambda_n \geq 0$: pairs of λ 's = $ib, -ib$.

Problem Set 6.5, page 350

- 3 Positive definite for $-3 < b < 3$ $\begin{bmatrix} 1 & 0 \\ b & 1 \end{bmatrix} \begin{bmatrix} 1 & b \\ 0 & 9-b^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ b & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 9-b^2 \end{bmatrix} \begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix} = LDL^T$
Positive definite for $c > 8$ $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 0 & c-8 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & c-8 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = LDL^T$.
- 4 $f(x, y) = x^2 + 4xy + 9y^2 = (x + 2y)^2 + 5y^2$; $x^2 + 6xy + 9y^2 = (x + 3y)^2$.
- 8 $A = \begin{bmatrix} 3 & 6 \\ 6 & 16 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$. Pivots 3, 4 outside squares, ℓ_{ij} inside. $x^T Ax = 3(x + 2y)^2 + 4y^2$
- 10 $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ has pivots $2, \frac{3}{2}, \frac{4}{3}$; $B = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$ is singular; $B \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.
- 12 A is positive definite for $c > 1$; determinants $c, c^2 - 1, (c - 1)^2(c + 2) > 0$. B is never positive definite (determinants $d - 4$ and $-4d + 12$ are never both positive).
- 14 The eigenvalues of A^{-1} are positive because they are $1/\lambda(A)$. And the entries of A^{-1} pass the determinant tests. And $x^T A^{-1} x = (A^{-1}x)^T A(A^{-1}x) > 0$ for all $x \neq 0$.
- 17 If a_{jj} were smaller than all λ 's, $A - a_{jj}I$ would have all eigenvalues > 0 (positive definite). But $A - a_{jj}I$ has a zero in the (j, j) position; impossible by Problem 16.
- 21 A is positive definite when $s > 8$; B is positive definite when $t > 5$ by determinants.
- 22 $R = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{9} & \\ & \sqrt{1} \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$; $R = Q \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix} Q^T = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$.
- 24 The ellipse $x^2 + xy + y^2 = 1$ has axes with half-lengths $1/\sqrt{\lambda} = \sqrt{2}$ and $\sqrt{2/3}$.
- 25 $A = C^T C = \begin{bmatrix} 9 & 3 \\ 3 & 5 \end{bmatrix}; \begin{bmatrix} 4 & 8 \\ 8 & 25 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $C = \begin{bmatrix} 2 & 4 \\ 0 & 3 \end{bmatrix}$
- 29 $H_1 = \begin{bmatrix} 6x^2 & 2x \\ 2x & 2 \end{bmatrix}$ is positive definite if $x \neq 0$; $F_1 = (\frac{1}{2}x^2 + y)^2 = 0$ on the curve $\frac{1}{2}x^2 + y = 0$; $H_2 = \begin{bmatrix} 6x & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is indefinite, $(0, 1)$ is a saddle point of F_2 .
- 31 If $c > 9$ the graph of z is a bowl, if $c < 9$ the graph has a saddle point. When $c = 9$ the graph of $z = (2x + 3y)^2$ is a "trough" staying at zero on the line $2x + 3y = 0$.
- 32 Orthogonal matrices, exponentials e^{At} , matrices with $\det = 1$ are groups. Examples of subgroups are orthogonal matrices with $\det = 1$, exponentials e^{An} for integer n .
- 34 The five eigenvalues of K are $2 - 2 \cos \frac{k\pi}{6} = 2 - \sqrt{3}, 2 - 1, 2, 2 + 1, 2 + \sqrt{3}$: product of eigenvalues = $6 = \det K$.

Problem Set 6.6, page 360

- 1 $B = GCG^{-1} = GF^{-1}AFG^{-1}$ so $M = FG^{-1}$. C similar to A and $B \Rightarrow A$ similar to B .
- 6 Eight families of similar matrices: six matrices have $\lambda = 0, 1$ (one family); three matrices have $\lambda = 1, 1$ and three have $\lambda = 0, 0$ (two families each!); one has $\lambda = 1, -1$; one has $\lambda = 2, 0$; two have $\lambda = \frac{1}{2}(1 \pm \sqrt{5})$ (they are in one family).
- 7 (a) $(M^{-1}AM)(M^{-1}x) = M^{-1}(Ax) = M^{-1}0 = 0$ (b) The nullspaces of A and of $M^{-1}AM$ have the same *dimension*. Different vectors and different bases.
- 8 Same Λ But $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$ have the same line of eigenvectors
Same S and the same eigenvalues $\lambda = 0, 0$.
- 10 $J^2 = \begin{bmatrix} c^2 & 2c \\ 0 & c^2 \end{bmatrix}$ and $J^k = \begin{bmatrix} c^k & kc^{k-1} \\ 0 & c^k \end{bmatrix}$; $J^0 = I$ and $J^{-1} = \begin{bmatrix} c^{-1} & -c^{-2} \\ 0 & c^{-1} \end{bmatrix}$.
- 14 (1) Choose M_i = reverse diagonal matrix to get $M_i^{-1}J_iM_i = M_i^T$ in each block
(2) M_0 has those diagonal blocks M_i to get $M_0^{-1}JM_0 = J^T$. (3) $A^T = (M^{-1})^T J^T M^T$ equals $(M^{-1})^T M_0^{-1}JM_0M^T = (MM_0M^T)^{-1}A(MM_0M^T)$, and A^T is similar to A .
- 17 (a) *False*: Diagonalize a nonsymmetric $A = S\Lambda S^{-1}$. Then Λ is symmetric and similar
(b) *True*: A singular matrix has $\lambda = 0$. (c) *False*: $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ are similar
(they have $\lambda = \pm 1$) (d) *True*: Adding I increases all eigenvalues by 1
- 18 $AB = B^{-1}(BA)B$ so AB is similar to BA . If $ABx = \lambda x$ then $BA(Bx) = \lambda(Bx)$.
- 19 Diagonal blocks 6 by 6, 4 by 4; AB has the same eigenvalues as BA plus 6 – 4 zeros.
- 22 $A = MJM^{-1}$, $A^n = MJ^nM^{-1} = 0$ (each J^k has 1's on the k th diagonal).
 $\det(A - \lambda I) = \lambda^n$ so $J^n = 0$ by the Cayley-Hamilton Theorem.

Problem Set 6.7, page 371

- 1 $A = U\Sigma V^T = \begin{bmatrix} u_1 & u_2 \end{bmatrix} \begin{bmatrix} \sigma_1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 & v_2 \end{bmatrix}^T = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 & 3 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} \sqrt{50} & 0 \\ 0 & 0 \end{bmatrix} \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$
- 4 $A^T A = AA^T = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ has eigenvalues $\sigma_1^2 = \frac{3 + \sqrt{5}}{2}$, $\sigma_2^2 = \frac{3 - \sqrt{5}}{2}$. But A is indefinite
 $\sigma_1 = (1 + \sqrt{5})/2 = \lambda_1(A)$, $\sigma_2 = (\sqrt{5} - 1)/2 = -\lambda_2(A)$; $u_1 = v_1$ but $u_2 = -v_2$.
- 5 A proof that *eigshow* finds the SVD. When $V_1 = (1, 0)$, $V_2 = (0, 1)$ the demo finds AV_1 and AV_2 at some angle θ . A 90° turn by the mouse to $V_2, -V_1$ finds AV_2 and $-AV_1$ at the angle $\pi - \theta$. Somewhere between, the constantly orthogonal v_1 and v_2 must produce Av_1 and Av_2 at angle $\pi/2$. Those orthogonal directions give u_1 and u_2 .
- 9 $A = UV^T$ since all $\sigma_j = 1$, which means that $\Sigma = I$.
- 14 The smallest change in A is to set its smallest singular value σ_2 to zero.
- 15 The singular values of $A + I$ are *not* $\sigma_j + 1$. Need eigenvalues of $(A + I)^T(A + I)$.
- 17 $A = U\Sigma V^T = [\text{cosines including } u_4] \text{diag}(\text{sqrt}(2 - \sqrt{2}), 2, 2 + \sqrt{2}) [\text{sine matrix}]^T$.
 $AV = U\Sigma$ says that differences of sines in V are cosines in U times σ 's.

Problem Set 7.1, page 380

- 3 $T(v) = (0, 1)$ and $T(v) = v_1 v_2$ are not linear.
- 4 (a) $S(T(v)) = v$ (b) $S(T(v_1) + T(v_2)) = S(T(v_1)) + S(T(v_2))$.
- 5 Choose $v = (1, 1)$ and $w = (-1, 0)$. $T(v) + T(w) = (0, 1)$ but $T(v + w) = (0, 0)$.
- 7 (a) $T(T(v)) = v$ (b) $T(T(v)) = v + (2, 2)$ (c) $T(T(v)) = -v$ (d) $T(T(v)) = T(v)$.
- 10 Not invertible: (a) $T(1, 0) = \mathbf{0}$ (b) $(0, 0, 1)$ is not in the range (c) $T(0, 1) = \mathbf{0}$.
- 12 Write v as a combination $c(1, 1) + d(2, 0)$. Then $T(v) = c(2, 2) + d(0, 0)$. $T(v) = (4, 4); (2, 2); (2, 2)$; if $v = (a, b) = b(1, 1) + \frac{a-b}{2}(2, 0)$ then $T(v) = b(2, 2) + (0, 0)$.
- 16 No matrix A gives $A \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. To professors: Linear transformations on matrix space come from 4 by 4 matrices. Those in Problems 13–15 were special.
- 17 (a) True (b) True (c) True (d) False.
- 19 $T(T^{-1}(M)) = M$ so $T^{-1}(M) = A^{-1}MB^{-1}$.
- 20 (a) Horizontal lines stay horizontal, vertical lines stay vertical (b) House squashes onto a line (c) Vertical lines stay vertical because $T(1, 0) = (a_{11}, 0)$.
- 27 Also 30 emphasizes that circles are transformed to ellipses (see figure in Section 6.7).
- 29 (a) $ad - bc = 0$ (b) $ad - bc > 0$ (c) $|ad - bc| = 1$. If vectors to two corners transform to themselves then by linearity $T = I$. (Fails if one corner is $(0, 0)$.)

Problem Set 7.2, page 395

- 3 (Matrix A)² = B when (transformation T)² = S and output basis = input basis.
- 5 $T(v_1 + v_2 + v_3) = 2w_1 + w_2 + 2w_3$; A times $(1, 1, 1)$ gives $(2, 1, 2)$.
- 6 $v = c(v_2 - v_3)$ gives $T(v) = \mathbf{0}$; nullspace is $(0, c, -c)$; solutions $(1, 0, 0) + (0, c, -c)$.
- 8 For $T^2(v)$ we would need to know $T(w)$. If the w 's equal the v 's, the matrix is A^2 .
- 12 (c) is wrong because w_1 is not generally in the input space.
- 14 (a) $\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$ (b) $\begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$ = inverse of (a) (c) $A \begin{bmatrix} 2 \\ 6 \end{bmatrix}$ must be $2A \begin{bmatrix} 1 \\ 3 \end{bmatrix}$.
- 16 $MN = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}^{-1} = \begin{bmatrix} 3 & -1 \\ -7 & 3 \end{bmatrix}$.
- 18 $(a, b) = (\cos \theta, -\sin \theta)$. Minus sign from $Q^{-1} = Q^T$.
- 20 $w_2(x) = 1 - x^2$; $w_3(x) = \frac{1}{2}(x^2 - x)$; $y = 4w_1 + 5w_2 + 6w_3$.
- 23 The matrix M with these nine entries must be invertible.
- 27 If T is not invertible, $T(v_1), \dots, T(v_n)$ is not a basis. We couldn't choose $w_i = T(v_i)$.
- 30 S takes (x, y) to $(-x, y)$. $S(T(v)) = (-1, 2)$. $S(v) = (-2, 1)$ and $T(S(v)) = (1, -2)$.
- 34 The last step writes 6, 6, 2, 2 as the overall average 4, 4, 4, 4 plus the difference 2, 2, -2, -2. Therefore $c_1 = 4$ and $c_2 = 2$ and $c_3 = 1$ and $c_4 = 1$.

- 35 The wavelet basis is $(1, 1, 1, 1, 1, 1, 1, 1)$ and the long wavelet and two medium wavelets $(1, 1, -1, -1, 0, 0, 0, 0)$, $(0, 0, 0, 0, 1, 1, -1, -1)$ and 4 wavelets with a single pair $1, -1$.
- 36 If $Vb = Wc$ then $b = V^{-1}Wc$. The change of basis matrix is $V^{-1}W$.
- 37 Multiplication by $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with this basis is represented by 4 by 4 $A = \begin{bmatrix} aI & bI \\ cI & dI \end{bmatrix}$
- 38 If $w_1 = Av_1$ and $w_2 = Av_2$ then $a_{11} = a_{22} = 1$. All other entries will be zero.

Problem Set 7.3, page 406

- 1 $A^T A = \begin{bmatrix} 10 & 20 \\ 20 & 40 \end{bmatrix}$ has $\lambda = 50$ and 0, $v_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $v_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ -1 \end{bmatrix}$; $\sigma_1 = \sqrt{50}$.
 $Av_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 5 \\ 15 \end{bmatrix} = \sigma_1 u_1$ and $Av_2 = \mathbf{0}$. $u_1 = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $AA^T u_1 = 50 u_1$.
- 3 $A = QH = \frac{1}{\sqrt{50}} \begin{bmatrix} 7 & -1 \\ 1 & 7 \end{bmatrix} \frac{1}{\sqrt{50}} \begin{bmatrix} 10 & 20 \\ 20 & 40 \end{bmatrix}$. H is semidefinite because A is singular.
- 4 $A^+ = V \begin{bmatrix} 1/\sqrt{50} & 0 \\ 0 & 0 \end{bmatrix} U^T = \frac{1}{50} \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}$; $A^+ A = \begin{bmatrix} .2 & .4 \\ .4 & .8 \end{bmatrix}$, $AA^+ = \begin{bmatrix} .1 & .3 \\ .3 & .9 \end{bmatrix}$.
- 7 $[\sigma_1 u_1 \quad \sigma_2 u_2] \begin{bmatrix} v_1^T \\ v_2^T \end{bmatrix} = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T$. In general this is $\sigma_1 u_1 v_1^T + \cdots + \sigma_r u_r v_r^T$.
- 9 A^+ is A^{-1} because A is invertible. Pseudoinverse equals inverse when A^{-1} exists!
- 11 $A = [1] [5 \ 0 \ 0] V^T$ and $A^+ = V \begin{bmatrix} .2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} .12 \\ .16 \\ 0 \end{bmatrix}$; $A^+ A = \begin{bmatrix} .36 & .48 & 0 \\ .48 & .64 & 0 \\ 0 & 0 & 0 \end{bmatrix}$; $AA^+ = [1]$
- 13 If $\det A = 0$ then $\text{rank}(A) < n$; thus $\text{rank}(A^+) < n$ and $\det A^+ = 0$.
- 16 x^+ in the row space of A is perpendicular to $\hat{x} - x^+$ in the nullspace of $A^T A =$ nullspace of A . The right triangle has $c^2 = a^2 + b^2$.
- 17 $AA^+ p = p$, $AA^+ e = \mathbf{0}$, $A^+ Ax_r = x_r$, $A^+ Ax_n = \mathbf{0}$.
- 19 L is determined by ℓ_{21} . Each eigenvector in S is determined by one number. The counts are 1 + 3 for LU , 1 + 2 + 1 for LDU , 1 + 3 for QR , 1 + 2 + 1 for $U\Sigma V^T$, 2 + 2 + 0 for $S\Lambda S^{-1}$.
- 22 Keep only the r by r corner Σ_r of Σ (the rest is all zero). Then $A = U\Sigma V^T$ has the required form $A = \hat{U} M_1 \Sigma_r M_2^T \hat{V}^T$ with an invertible $M = M_1 \Sigma_r M_2^T$ in the middle.
- 23 $\begin{bmatrix} 0 & A \\ A^T & 0 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} Av \\ A^T u \end{bmatrix} = \sigma \begin{bmatrix} u \\ v \end{bmatrix}$. The singular values of A are eigenvalues of this block matrix.

Problem Set 8.1, page 418

- 3 The rows of the free-free matrix in equation (9) add to $[0 \ 0 \ 0]$ so the right side needs $f_1 + f_2 + f_3 = 0$. $f = (-1, 0, 1)$ gives $c_2 u_1 - c_2 u_2 = -1$, $c_3 u_2 - c_3 u_3 = -1$, $0 = 0$. Then $u_{\text{particular}} = (-c_2^{-1} - c_3^{-1}, -c_3^{-1}, 0)$. Add any multiple of $u_{\text{nullspace}} = (1, 1, 1)$.
- 4 $\int -\frac{d}{dx} \left(c(x) \frac{du}{dx} \right) dx = - \left[c(x) \frac{du}{dx} \right]_0^1 = 0$ (bdry cond) so we need $\int f(x) dx = 0$.
- 6 Multiply $A_1^T C_1 A_1$ as columns of A_1^T times c 's times rows of A_1 . The first 3 by 3 "element matrix" $c_1 E_1 = [1 \ 0 \ 0]^T c_1 [1 \ 0 \ 0]$ has c_1 in the top left corner.
- 8 The solution to $-u'' = 1$ with $u(0) = u(1) = 0$ is $u(x) = \frac{1}{2}(x - x^2)$. At $x = \frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}$ this gives $u = 2, 3, 3, 2$ (discrete solution in Problem 7) times $(\Delta x)^2 = 1/25$.
- 11 Forward/backward/centered for du/dx has a big effect because that term has the large coefficient. MATLAB: $E = \text{diag}(\text{ones}(6, 1), 1)$; $K = 64 * (2 * \text{eye}(7) - E - E')$; $D = 80 * (E - \text{eye}(7))$; $(K + D) \backslash \text{ones}(7, 1)$; % forward; $(K - D') \backslash \text{ones}(7, 1)$; % backward; $(K + D/2 - D'/2) \backslash \text{ones}(7, 1)$; % centered is usually the best: more accurate

Problem Set 8.2, page 428

- 1 $A = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix}$; nullspace contains $\begin{bmatrix} c \\ c \\ c \end{bmatrix}$; $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ is not orthogonal to that nullspace.
- 2 $A^T y = 0$ for $y = (1, -1, 1)$; current along edge 1, edge 3, back on edge 2 (full loop).
- 5 Kirchhoff's Current Law $A^T y = f$ is solvable for $f = (1, -1, 0)$ and not solvable for $f = (1, 0, 0)$; f must be orthogonal to $(1, 1, 1)$ in the nullspace: $f_1 + f_2 + f_3 = 0$.
- 6 $A^T A x = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} x = \begin{bmatrix} 3 \\ -3 \\ 0 \end{bmatrix} = f$ produces $x = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + \begin{bmatrix} c \\ c \\ c \end{bmatrix}$; potentials $x = 1, -1, 0$ and currents $-Ax = 2, 1, -1$; f sends 3 units from node 2 into node 1.
- 7 $A^T \begin{bmatrix} 1 & & \\ & 2 & \\ & & 2 \end{bmatrix} A = \begin{bmatrix} 3 & -1 & -2 \\ -1 & 3 & -2 \\ -2 & -2 & 4 \end{bmatrix}$; $f = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ yields $x = \begin{bmatrix} 5/4 \\ 1 \\ 7/8 \end{bmatrix} + \text{any } \begin{bmatrix} c \\ c \\ c \end{bmatrix}$; potentials $x = \frac{5}{4}, 1, \frac{7}{8}$ and currents $-CAx = \frac{1}{4}, \frac{3}{4}, \frac{1}{4}$.
- 9 Elimination on $Ax = b$ always leads to $y^T b = 0$ in the zero rows of U and R : $-b_1 + b_2 - b_3 = 0$ and $b_3 - b_4 + b_5 = 0$ (those y 's are from Problem 8 in the left nullspace). This is Kirchhoff's Voltage Law around the two loops.
- 11 $A^T A = \begin{bmatrix} 2 & -1 & -1 & 0 \\ -1 & 3 & -1 & -1 \\ -1 & -1 & 3 & -1 \\ 0 & -1 & -1 & 2 \end{bmatrix}$ diagonal entry = number of edges into the node
the trace is 2 times the number of nodes
off-diagonal entry = -1 if nodes are connected
 $A^T A$ is the **graph Laplacian**, $A^T C A$ is **weighted** by C
- 13 $A^T C A x = \begin{bmatrix} 4 & -2 & -2 & 0 \\ -2 & 8 & -3 & -3 \\ -2 & -3 & 8 & -3 \\ 0 & -3 & -3 & 6 \end{bmatrix} x = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}$ gives four potentials $x = (\frac{5}{12}, \frac{1}{6}, \frac{1}{6}, 0)$
I grounded $x_4 = 0$ and solved for x
currents $y = -CAx = (\frac{2}{3}, \frac{2}{3}, 0, \frac{1}{2}, \frac{1}{2})$

- 17 (a) 8 independent columns (b) f must be orthogonal to the nullspace so f 's add to zero (c) Each edge goes into 2 nodes, 12 edges make diagonal entries sum to 24.

Problem Set 8.3, page 437

- 2 $A = \begin{bmatrix} .6 & -1 \\ .4 & 1 \end{bmatrix} \begin{bmatrix} 1 & .75 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -.4 & .6 \end{bmatrix}; A^\infty = \begin{bmatrix} .6 & -1 \\ .4 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -.4 & .6 \end{bmatrix} = \begin{bmatrix} .6 & .6 \\ .4 & .4 \end{bmatrix}.$
- 3 $\lambda = 1$ and $.8$, $x = (1, 0)$; 1 and $-.8$, $x = (\frac{5}{9}, \frac{4}{9})$; $1, \frac{1}{4}$, and $\frac{1}{4}$, $x = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$.
- 5 The steady state eigenvector for $\lambda = 1$ is $(0, 0, 1)$ = everyone is dead.
- 6 Add the components of $Ax = \lambda x$ to find sum $s = \lambda s$. If $\lambda \neq 1$ the sum must be $s = 0$.
- 7 $(.5)^k \rightarrow 0$ gives $A^k \rightarrow A^\infty$; any $A = \begin{bmatrix} .6 + .4a & .6 - .6a \\ .4 - .4a & .4 + .6a \end{bmatrix}$ with $\begin{matrix} a \leq 1 \\ .4 + .6a \geq 0 \end{matrix}$
- 9 M^2 is still nonnegative; $[1 \ \cdots \ 1]M = [1 \ \cdots \ 1]$ so multiply on the right by M to find $[1 \ \cdots \ 1]M^2 = [1 \ \cdots \ 1] \Rightarrow$ columns of M^2 add to 1.
- 10 $\lambda = 1$ and $a + d - 1$ from the trace; steady state is a multiple of $x_1 = (b, 1 - a)$.
- 12 B has $\lambda = 0$ and $-.5$ with $x_1 = (.3, .2)$ and $x_2 = (-1, 1)$; A has $\lambda = 1$ so $A - I$ has $\lambda = 0$. $e^{-.5t}$ approaches zero and the solution approaches $c_1 e^{0t} x_1 = c_1 x_1$.
- 13 $x = (1, 1, 1)$ is an eigenvector when the row sums are equal; $Ax = (.9, .9, .9)$.
- 15 The first two A 's have $\lambda_{\max} < 1$; $p = \begin{bmatrix} 8 \\ 6 \end{bmatrix}$ and $\begin{bmatrix} 130 \\ 32 \end{bmatrix}$; $I - \begin{bmatrix} .5 & 1 \\ .5 & 0 \end{bmatrix}$ has no inverse.
- 16 $\lambda = 1$ (Markov), 0 (singular), $.2$ (from trace). Steady state $(.3, .3, .4)$ and $(30, 30, 40)$.
- 17 No, A has an eigenvalue $\lambda = 1$ and $(I - A)^{-1}$ does not exist.
- 19 Λ times $S^{-1} \Delta S$ has the same diagonal as $S^{-1} \Delta S$ times Λ because Λ is diagonal.
- 20 If $B > A > 0$ and $Ax = \lambda_{\max}(A)x > 0$ then $Bx > \lambda_{\max}(A)x$ and $\lambda_{\max}(B) > \lambda_{\max}(A)$.

Problem Set 8.4, page 446

- 1 Feasible set = line segment $(6, 0)$ to $(0, 3)$; minimum cost at $(6, 0)$, maximum at $(0, 3)$.
- 2 Feasible set has corners $(0, 0)$, $(6, 0)$, $(2, 2)$, $(0, 6)$. Minimum cost $2x - y$ at $(6, 0)$.
- 3 Only two corners $(4, 0, 0)$ and $(0, 2, 0)$; let $x_i \rightarrow -\infty$, $x_2 = 0$, and $x_3 = x_1 - 4$.
- 4 From $(0, 0, 2)$ move to $x = (0, 1, 1.5)$ with the constraint $x_1 + x_2 + 2x_3 = 4$. The new cost is $3(1) + 8(1.5) = \$15$ so $r = -1$ is the reduced cost. The simplex method also checks $x = (1, 0, 1.5)$ with cost $5(1) + 8(1.5) = \$17$; $r = 1$ means more expensive.
- 5 $c = [3 \ 5 \ 7]$ has minimum cost 12 by the Ph.D. since $x = (4, 0, 0)$ is minimizing. The dual problem maximizes $4y$ subject to $y \leq 3$, $y \leq 5$, $y \leq 7$. Maximum = 12.
- 8 $y^T b \leq y^T Ax = (A^T y)^T x \leq c^T x$. The first inequality needed $y \geq 0$ and $Ax - b \geq 0$.

Problem Set 8.5, page 451

- 1 $\int_0^{2\pi} \cos((j+k)x) dx = \left[\frac{\sin((j+k)x)}{j+k} \right]_0^{2\pi} = 0$ and similarly $\int_0^{2\pi} \cos((j-k)x) dx = 0$
Notice $j-k \neq 0$ in the denominator. If $j=k$ then $\int_0^{2\pi} \cos^2 jx dx = \pi$.
- 4 $\int_{-1}^1 (1)(x^3 - cx) dx = 0$ and $\int_{-1}^1 (x^2 - \frac{1}{3})(x^3 - cx) dx = 0$ for all c (odd functions).
Choose c so that $\int_{-1}^1 x(x^3 - cx) dx = [\frac{1}{5}x^5 - \frac{c}{3}x^3]_{-1}^1 = \frac{2}{5} - c\frac{2}{3} = 0$. Then $c = \frac{3}{5}$.
- 5 The integrals lead to the Fourier coefficients $a_1 = 0$, $b_1 = 4/\pi$, $b_2 = 0$.
- 6 From eqn. (3) $a_k = 0$ and $b_k = 4/\pi k$ (odd k). The square wave has $\|f\|^2 = 2\pi$.
Then eqn. (6) is $2\pi = \pi(16/\pi^2)(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots)$. That infinite series equals $\pi^2/8$.
- 8 $\|v\|^2 = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots = 2$ so $\|v\| = \sqrt{2}$; $\|v\|^2 = 1 + a^2 + a^4 + \dots = 1/(1-a^2)$
so $\|v\| = 1/\sqrt{1-a^2}$; $\int_0^{2\pi} (1 + 2\sin x + \sin^2 x) dx = 2\pi + 0 + \pi$ so $\|f\| = \sqrt{3\pi}$.
- 9 (a) $f(x) = (1 + \text{square wave})/2$ so the a 's are $\frac{1}{2}, 0, 0, \dots$ and the b 's are $2/\pi, 0, -2/3\pi, 0, 2/5\pi, \dots$ (b) $a_0 = \int_0^{2\pi} x dx / 2\pi = \pi$, all other $a_k = 0, b_k = -2/k$.
- 11 $\cos^2 x = \frac{1}{2} + \frac{1}{2} \cos 2x$; $\cos(x + \frac{\pi}{3}) = \cos x \cos \frac{\pi}{3} - \sin x \sin \frac{\pi}{3} = \frac{1}{2} \cos x - \frac{\sqrt{3}}{2} \sin x$.
- 13 $a_0 = \frac{1}{2\pi} \int F(x) dx = \frac{1}{2\pi}$, $a_k = \frac{\sin(kh/2)}{\pi kh/2} \rightarrow \frac{1}{\pi}$ for delta function; all $b_k = 0$.

Problem Set 8.6, page 458

- 3 If $\sigma_3 = 0$ the third equation is exact.
- 4 0, 1, 2 have probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$ and $\sigma^2 = (0-1)^2 \frac{1}{4} + (1-1)^2 \frac{1}{2} + (2-1)^2 \frac{1}{4} = \frac{1}{2}$.
- 5 Mean $(\frac{1}{2}, \frac{1}{2})$. Independent flips lead to $\Sigma = \text{diag}(\frac{1}{4}, \frac{1}{4})$. Trace = $\sigma_{\text{total}}^2 = \frac{1}{2}$.
- 6 Mean $m = p_0$ and variance $\sigma^2 = (1-p_0)^2 p_0 + (0-p_0)^2 (1-p_0) = p_0(1-p_0)$.
- 7 Minimize $P = a^2 \sigma_1^2 + (1-a)^2 \sigma_2^2$ at $P' = 2a\sigma_1^2 - 2(1-a)\sigma_2^2 = 0$; $a = \sigma_2^2 / (\sigma_1^2 + \sigma_2^2)$
recovers equation (2) for the statistically correct choice with minimum variance.
- 8 Multiply $L\Sigma L^T = (A^T \Sigma^{-1} A)^{-1} A^T \Sigma^{-1} \Sigma \Sigma^{-1} A (A^T \Sigma^{-1} A)^{-1} = P = (A^T \Sigma^{-1} A)^{-1}$.
- 9 Row 3 = -row 1 and row 4 = -row 2: A has rank 2.

Problem Set 8.7, page 464

- 1 (x, y, z) has homogeneous coordinates (cx, cy, cz, c) for $c = 1$ and all $c \neq 0$.
- 4 $S = \text{diag}(c, c, c, 1)$; row 4 of ST and TS is 1, 4, 3, 1 and $c, 4c, 3c, 1$; use $vTS!$
- 5 $S = \begin{bmatrix} 1/8.5 & & \\ & 1/11 & \\ & & 1 \end{bmatrix}$ for a 1 by 1 square, starting from an 8.5 by 11 page.
- 9 $n = \left(\frac{2}{3}, \frac{2}{3}, \frac{1}{3}\right)$ has $P = I - nn^T = \frac{1}{9} \begin{bmatrix} 5 & -4 & -2 \\ -4 & 5 & -2 \\ -2 & -2 & 8 \end{bmatrix}$. Notice $\|n\| = 1$.

- 10 We can choose $(0, 0, 3)$ on the plane and multiply $T_- P T_+ = \frac{1}{9} \begin{bmatrix} 5 & -4 & -2 & 0 \\ -4 & 5 & -2 & 0 \\ -2 & -2 & 8 & 0 \\ 6 & 6 & 3 & 9 \end{bmatrix}$.
- 11 $(3, 3, 3)$ projects to $\frac{1}{3}(-1, -1, 4)$ and $(3, 3, 3, 1)$ projects to $(\frac{1}{3}, \frac{1}{3}, \frac{5}{3}, 1)$. Row vectors!
- 13 That projection of a cube onto a plane produces a hexagon.
- 14 $(3, 3, 3)(I - 2nn^T) = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) \begin{bmatrix} 1 & -8 & -4 \\ -8 & 1 & -4 \\ -4 & -4 & 7 \end{bmatrix} = \left(-\frac{11}{3}, -\frac{11}{3}, -\frac{1}{3}\right)$.
- 15 $(3, 3, 3, 1) \rightarrow (3, 3, 0, 1) \rightarrow (-\frac{7}{3}, -\frac{7}{3}, -\frac{8}{3}, 1) \rightarrow (-\frac{7}{3}, -\frac{7}{3}, \frac{1}{3}, 1)$.
- 17 Space is rescaled by $1/c$ because (x, y, z, c) is the same point as $(x/c, y/c, z/c, 1)$.

Problem Set 9.1, page 472

- 1 Without exchange, pivots .001 and 1000; with exchange, 1 and -1 . When the pivot is larger than the entries below it, all $|\ell_{ij}| = |\text{entry/pivot}| \leq 1$. $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix}$.
- 4 The largest $\|x\| = \|A^{-1}b\|$ is $\|A^{-1}\| = 1/\lambda_{\min}$ since $A^T = A$; largest error $10^{-16}/\lambda_{\min}$.
- 5 Each row of U has at most w entries. Then w multiplications to substitute components of x (already known from below) and divide by the pivot. Total for n rows $< wn$.
- 6 The triangular L^{-1} , U^{-1} , R^{-1} need $\frac{1}{2}n^2$ multiplications. Q needs n^2 to multiply the right side by $Q^{-1} = Q^T$. So $QRx = b$ takes 1.5 times longer than $LUx = b$.
- 7 $UU^{-1} = I$: Back substitution needs $\frac{1}{2}j^2$ multiplications on column j , using the j by j upper left block. Then $\frac{1}{2}(1^2 + 2^2 + \cdots + n^2) \approx \frac{1}{2}(\frac{1}{3}n^3) = \text{total to find } U^{-1}$.
- 10 With 16-digit floating point arithmetic the errors $\|x - x_{\text{computed}}\|$ for $\varepsilon = 10^{-3}, 10^{-6}, 10^{-9}, 10^{-12}, 10^{-15}$ are of order $10^{-16}, 10^{-11}, 10^{-7}, 10^{-4}, 10^{-3}$.
- 11 (a) $\cos \theta = \frac{1}{\sqrt{10}}$, $\sin \theta = \frac{-3}{\sqrt{10}}$, $R = Q_{21}A = \frac{1}{\sqrt{10}} \begin{bmatrix} 10 & 14 \\ 0 & 8 \end{bmatrix}$ (b) $\lambda = 4$; use $-\theta$
 $x = (1, -3)/\sqrt{10}$
- 13 $Q_{ij}A$ uses $4n$ multiplications (2 for each entry in rows i and j). By factoring out $\cos \theta$, the entries 1 and $\pm \tan \theta$ need only $2n$ multiplications, which leads to $\frac{2}{3}n^3$ for QR .

Problem Set 9.2, page 478

- 1 $\|A\| = 2$, $\|A^{-1}\| = 2$, $c = 4$; $\|A\| = 3$, $\|A^{-1}\| = 1$, $c = 3$; $\|A\| = 2 + \sqrt{2} = \lambda_{\max}$ for positive definite A , $\|A^{-1}\| = 1/\lambda_{\min}$, $c = (2 + \sqrt{2})/(2 - \sqrt{2}) = 5.83$.
- 3 For the first inequality replace x by Bx in $\|Ax\| \leq \|A\|\|x\|$; the second inequality is just $\|Bx\| \leq \|B\|\|x\|$. Then $\|AB\| = \max(\|ABx\|/\|x\|) \leq \|A\|\|B\|$.
- 7 The triangle inequality gives $\|Ax + Bx\| \leq \|Ax\| + \|Bx\|$. Divide by $\|x\|$ and take the maximum over all nonzero vectors to find $\|A + B\| \leq \|A\| + \|B\|$.

- 8 If $Ax = \lambda x$ then $\|Ax\|/\|x\| = |\lambda|$ for that particular vector x . When we maximize the ratio over all vectors we get $\|A\| \geq |\lambda|$.
- 13 The residual $b - Ay = (10^{-7}, 0)$ is much smaller than $b - Az = (.0013, .0016)$. But z is much closer to the solution than y .
- 14 $\det A = 10^{-6}$ so $A^{-1} = 10^3 \begin{bmatrix} 659 & -563 \\ -913 & 780 \end{bmatrix}$: $\|A\| > 1$, $\|A^{-1}\| > 10^6$, then $c > 10^6$.
- 16 $x_1^2 + \cdots + x_n^2$ is not smaller than $\max(x_i^2)$ and not larger than $(|x_1| + \cdots + |x_n|)^2 = \|x\|_1^2$. $x_1^2 + \cdots + x_n^2 \leq n \max(x_i^2)$ so $\|x\| \leq \sqrt{n}\|x\|_\infty$. Choose $y_i = \text{sign } x_i = \pm 1$ to get $\|x\|_1 = x \cdot y \leq \|x\|\|y\| = \sqrt{n}\|x\|$. $x = (1, \dots, 1)$ has $\|x\|_1 = \sqrt{n}\|x\|$.

Problem Set 9.3, page 489

- 2 If $Ax = \lambda x$ then $(I - A)x = (1 - \lambda)x$. Real eigenvalues of $B = I - A$ have $|1 - \lambda| < 1$ provided λ is between 0 and 2.
- 6 Jacobi has $S^{-1}T = \frac{1}{3} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ with $|\lambda|_{\max} = \frac{1}{3}$. Small problem, fast convergence.
- 7 Gauss-Seidel has $S^{-1}T = \begin{bmatrix} 0 & \frac{1}{3} \\ 0 & \frac{1}{9} \end{bmatrix}$ with $|\lambda|_{\max} = \frac{1}{9}$ which is $(|\lambda|_{\max} \text{ for Jacobi})^2$.
- 9 Set the trace $2 - 2\omega + \frac{1}{4}\omega^2$ equal to $(\omega - 1) + (\omega - 1)$ to find $\omega_{\text{opt}} = 4(2 - \sqrt{3}) \approx 1.07$. The eigenvalues $\omega - 1$ are about .07, a big improvement.
- 15 In the j th component of Ax_1 , $\lambda_1 \sin \frac{j\pi}{n+1} = 2 \sin \frac{j\pi}{n+1} - \sin \frac{(j-1)\pi}{n+1} - \sin \frac{(j+1)\pi}{n+1}$. The last two terms combine into $-2 \sin \frac{j\pi}{n+1} \cos \frac{\pi}{n+1}$. Then $\lambda_1 = 2 - 2 \cos \frac{\pi}{n+1}$.
- 17 $A^{-1} = \frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ gives $u_1 = \frac{1}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $u_2 = \frac{1}{9} \begin{bmatrix} 5 \\ 4 \end{bmatrix}$, $u_3 = \frac{1}{27} \begin{bmatrix} 14 \\ 13 \end{bmatrix} \rightarrow u_\infty = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$.
- 18 $R = Q^T A = \begin{bmatrix} 1 & \cos \theta \sin \theta \\ 0 & -\sin^2 \theta \end{bmatrix}$ and $A_1 = RQ = \begin{bmatrix} \cos \theta(1 + \sin^2 \theta) & -\sin^3 \theta \\ -\sin^3 \theta & -\cos \theta \sin^2 \theta \end{bmatrix}$.
- 20 If $A - cI = QR$ then $A_1 = RQ + cI = Q^{-1}(QR + cI)Q = Q^{-1}AQ$. No change in eigenvalues because A_1 is similar to A .
- 21 Multiply $Aq_j = b_{j-1}q_{j-1} + a_jq_j + b_jq_{j+1}$ by q_j^T to find $q_j^T Aq_j = a_j$ (because the q 's are orthonormal). The matrix form (multiplying by columns) is $AQ = QT$ where T is *tridiagonal*. The entries down the diagonals of T are the a 's and b 's.
- 23 If A is symmetric then $A_1 = Q^{-1}AQ = Q^T AQ$ is also symmetric. $A_1 = RQ = R(QR)R^{-1} = RAR^{-1}$ has R and R^{-1} upper triangular, so A_1 cannot have nonzeros on a lower diagonal than A . If A is tridiagonal and symmetric then (by using symmetry for the upper part of A_1) the matrix $A_1 = RAR^{-1}$ is also tridiagonal.
- 26 If each center a_{ii} is larger than the circle radius r_i (this is diagonal dominance), then 0 is outside all circles: not an eigenvalue so A^{-1} exists.